DRDO-ORIENTED ACADEMIC PROJECT | B.TECH MECHANICAL ENGINEERING

Database-Driven Aircraft Engine Performance Analysis, Digital Twin, and Intelligent Engine Selection System

Project Definition and Architecture Document

DOCUMENT TYPE

Project Definition & Architecture Document

PROGRAMME

B.Tech Mechanical Engineering

VERSION

1.1 (Revised)

DATE

September 2026

SUPERSEDES

Version 1.0 (September 2026)

CLASSIFICATION

Academic — Public Domain Data Only

Table of Contents

1. Final Problem Statement 3
2. Objectives 4
3. Scope 6
4. Functional Requirements 7
5. Non-Functional Requirements 9
6. Engine Categories 10
7. Engine Selection Criteria 11
8. Engine Manufacturers 12
9. Engine Selection — Representative Engines 13
10. Database Architecture 18
11. Database Schema 20
12. Primary and Foreign Keys 24
13. Entity-Relationship Diagram 26
14. Data Fields Summary 28
15. Data Sources 30
16. Atmospheric Model (ISA) 31
17. Engine Modelling Methodology 33
18. Model Validity Checks 40
19. Digital Twin Architecture 44
20. Machine Learning Architecture 46
21. Engine Selection Optimisation 50
22. GUI Architecture 52
23. Validation Methodology 55
24. Testing Strategy 58
25. Project Milestones 59
26. Risks and Limitations 62
27. Expected Outputs 65
28. Version 1.1 Change Log 66
29. Implementation Freeze Checklist 73

1. Final Problem Statement

1.1 Background

Aircraft gas turbine engines are among the most thermodynamically complex machines produced by human engineering. They operate across extraordinary ranges of altitude (sea level to ~20 km), Mach number (0 to >2.5), throttle setting, and ambient condition, producing net thrust through fundamentally different cycle arrangements depending on their architecture. The global inventory of airworthy gas turbine engines encompasses dozens of distinct design families — turbojets, turbofans (low-bypass military and high-bypass commercial), turboprops, and turboshafts — produced by manufacturers on multiple continents under radically different design philosophies, operational constraints, and national priorities.

Each engine architecture presents a distinct thermodynamic cycle, a characteristic performance envelope, and a unique set of trade-offs between thrust, specific fuel consumption, weight, reliability, altitude capability, and cost of ownership. For any aerospace systems engineer, defence procurement authority, or academic researcher, the challenge is not merely to characterise an individual engine but to evaluate candidate engines relative to one another and relative to a specific mission requirement. This is a fundamentally multi-dimensional optimisation problem: an engine that delivers superior sea-level thrust may be substantially outperformed at 15 km altitude, and one optimised for high-subsonic cruise is ill-suited to supersonic dash.

In the Indian defence aviation context, this challenge is acutely relevant. The Indian Air Force (IAF) operates engines of Soviet, French, British, American, and domestic origin simultaneously. The Defence Research and Development Organisation (DRDO) programme to develop the GTX-35VS Kaveri turbofan represents India's first concerted attempt to achieve engine self-sufficiency — making the ability to analyse, compare, and predict engine performance not merely an academic exercise but a matter of strategic importance. No publicly available, open-source, MATLAB-integrated tool currently enables systematic comparison of such a diverse engine portfolio with full provenance tracking of every numerical parameter.

1.2 Core Engineering Problem

The core engineering problem addressed by this project is the absence of a unified, academically rigorous, publicly documented framework that can: (a) store and retrieve verified engine specifications with explicit source traceability for every parameter; (b) execute thermodynamic cycle analysis for three principal engine architectures from a common interface; (c) construct an academic steady-state software Digital Twin that connects physics-model outputs to a relational database; (d) train and evaluate ML surrogate models as speed-optimised replacements for iterative physical simulation; and (e) apply these capabilities in a structured mission-based engine selection methodology.

The absence of such a framework means that currently, engine comparisons in academic literature are typically performed in isolation — one engine per paper, with limited cross-architecture consistency. Performance estimates for restricted engines like the Kaveri are often presented without clarity on whether values are manufacturer-confirmed, government-published, academically estimated, or simply assumed. This ambiguity has engineering consequences: design decisions based on unreliable reference data produce unreliable conclusions.

1.3 Proposed Solution

This project constructs a MATLAB + SQLite integrated system that addresses all five deficiencies above within the constraints of a B.Tech academic project, using only public-domain data and standard educational toolboxes. The system explicitly separates Reference data (from published sources), Simulated data (from the project's physics models), and ML_Predicted data (from trained surrogates) — never mixing these provenance classes during validation. Every numerical parameter stored in the database is classified by source type and confidence level, enabling the user to immediately distinguish manufacturer-confirmed values from academic estimates.

> **v1.1 Revision Note:** This document supersedes Version 1.0 (September 2026). All 20 correction areas identified in the v1.0 review have been addressed. Key changes: turboprop selection corrected (TV3-117VM turboshaft replaced with PW127G genuine turboprop); thermodynamic equations revised for proper choked/unchoked nozzle treatment; database schema extended with a Parameter_Provenance table for per-value source traceability; ML validation methodology separated into physics-validation and surrogate-validation streams; project milestones reordered so ML training begins only after physics model validation. Full change details are in Section 28 (Version 1.1 Change Log).

2. Objectives

The following ten objectives define the measurable outcomes of the project. Each objective carries an explicit validation criterion used in Section 23 to verify achievement.

Table 2.1 Project Objectives and Validation Criteria

ID	Objective	Measurable Outcome	Validation Criterion

OBJ-01	Build a normalised SQLite database with verified engine specifications for 11 Phase 1 engines with per-parameter source traceability.	Database contains all 11 engines; every parameter in Engine_Thermo_Params has a corresponding Parameter_Provenance record identifying source type and confidence.	SQL query confirms Parameter_Provenance rows ≥ number of populated Thermo_Params rows; all Confidence values are populated.
OBJ-02	Implement and validate an ISA atmospheric model covering 0–32 km.	ISA model reproduces ICAO Doc 7488/3 reference values at eight standard altitudes to within ±0.1%.	Unit test comparing MATLAB output against Table 16.3 values with tolerance check.
OBJ-03	Implement and validate a steady-state turbojet thermodynamic model for three turbojet engines.	Thrust prediction within ±10% of published/estimated reference for J79 and Atar 9C (High-confidence sources); within ±20% for R-25 (Medium-confidence sources) at stated reference conditions.	Section 23 validation table with computed vs reference for each engine.
OBJ-04	Implement and validate a separate-exhaust turbofan model for five turbofan engines spanning BPR 0.16–5.4.	Thrust and TSFC predictions within ±10% of published reference for High-confidence engines (CFM56-5B4, CF6-80C2); within ±20% for Medium-confidence engines (F100-PW-229, M88-2); within ±30% for Low-confidence engines (Kaveri — estimated values only).	Section 23 validation table; Kaveri results carry "Estimated Academic Model" disclaimer.
OBJ-05	Implement and validate a turboprop model for three turboprop engines.	Shaft power and BSFC predictions within ±10% of published reference for PT6A-67D and PW127G; within ±15% for AE 2100D3.	Section 23 validation table.
OBJ-06	Construct a five-level academic steady-state software Digital Twin that persists simulation data in the database.	All five DT levels (L1–L5) implemented and functional; performance data written to Performance_Data with correct Data_Type = 'Simulated'.	Integration test: run DT L4 for each engine; verify DB contains new Simulated records with valid foreign keys.
OBJ-07	Train and evaluate engine-architecture-specific ML surrogate models for all three architecture types.	Best surrogate model achieves $R^2 \geq 0.95$ on held-out physics-model data for each target variable. ($R^2 \geq 0.99$ is the aspirational target; actual results are reported as-found.)	Test-set evaluation logged; results compared against physics model predictions — NOT against real engine measurements.

OBJ-08	Implement a mission-based engine selection algorithm with hard-constraint filtering followed by weighted scoring.	Algorithm correctly excludes constraint-violating engines before scoring; handles empty feasible set; weighted scores sum correctly.	Unit test with deliberate constraint violations verifying exclusion; weight-sum assertion in code.
OBJ-09	Develop an eight-tab MATLAB App Designer GUI displaying units, model type, data confidence, and appropriate warnings.	GUI functional with all eight tabs; Kaveri results display "ESTIMATED / ACADEMIC MODEL" warning; all numeric outputs labelled with units and model type.	GUI walkthrough test covering all tabs; screenshot evidence of warning display.
OBJ-10	Document the complete project in a final technical report and a GitHub repository with ≥20 commits.	Final report ≥50 pages covering all objectives; repository has structured directory tree, README, and ≥20 commits.	Repository commit log; page count of final report PDF.

3. Scope

3.1 In Scope — Phase 1 (This Project)

- Steady-state thermodynamic cycle analysis for three gas turbine architectures: turbojet, turbofan (separate-exhaust), turboprop
- Academic steady-state software Digital Twin (L1–L5) for 11 representative engines (see Section 9)
- SQLite relational database with per-parameter source traceability via Parameter_Provenance table
- ISA atmospheric model (0–32 km geometric altitude)
- ML surrogate training on physics-model-generated data; architecture-specific models
- Weighted mission-based engine selection with hard-constraint pre-filtering
- MATLAB App Designer GUI (8 tabs) with units, model type, and confidence warnings
- Validation against published reference data (where available with stated confidence)
- Public-domain data only; all values source-classified
- MATLAB R2023b + Database Toolbox + Statistics and Machine Learning Toolbox + Simulink
- GitHub repository with documentation

3.2 Explicitly Out of Scope — Phase 1

- Transient engine dynamics and time-domain simulation
- Compressor and turbine characteristic maps (blade-level modelling)
- Surge, stall, and rotating stall modelling

- Variable geometry scheduling (VGV, VSV)
- Afterburner / reheater thermochemistry (no afterburner-on simulation in Phase 1)
- Turboshaft engines (Phase 2)
- Ramjet / scramjet / combined-cycle engines (Phase 3)
- Real hardware-in-the-loop or real-time embedded implementation
- Certified or type-approved performance prediction
- Classified, ITAR-restricted, or proprietary data
- Aircraft drag, aircraft weight, or mission range calculation
- Multi-engine aircraft configurations
- Engine degradation and health monitoring
- Inlet distortion and installation effects
- Bleed air and power offtakes
- Detailed combustion chemistry or emissions modelling

3.3 Data Policy Statement

> **DO NOT FABRICATE DATA.** Every numerical value in this document and in the project database must trace to a verifiable public-domain source. Where a parameter cannot be verified from a reliable source, it is explicitly marked with one of: *Manufacturer-Confirmed*, *Government/TC Source*, *Academic Literature*, *Calculated*, *Estimated*, *Assumed*, *Unavailable*. No estimated or assumed value may be presented as manufacturer-confirmed. For the Kaveri/GTX-35VS engine, all parameters are Low confidence and all model outputs carry the disclaimer "ESTIMATED / ACADEMIC MODEL — NOT MANUFACTURER-CONFIRMED PERFORMANCE."

4. Functional Requirements

Functional requirements define specific system behaviours. Each is given an identifier, priority (P1 = mandatory, P2 = important, P3 = desirable), and a reference to the implementing MATLAB module.

FR-01 The system shall connect to the SQLite database `engine_performance.db` using the MATLAB Database Toolbox JDBC driver and enforce referential integrity via `PRAGMA foreign_keys = ON`.

FR-02 The system shall populate the database with specifications for 11 Phase 1 engines. Every numerical parameter shall have a corresponding record in the Parameter_Provenance table specifying Source_Type and Confidence_Level.

FR-03 The system shall implement an ISA atmospheric model (0–32 km geometric altitude) returning T, P, ρ, a, and μ at any specified altitude, with stagnation conditions at any specified Mach number.

FR-04 The system shall implement a steady-state turbojet thermodynamic model using stations 0–2–3–4–5–8, with correct stagnation properties, compressor work, combustor energy balance, turbine power balance, and choked/unchoked nozzle treatment.

FR-05 The system shall implement a steady-state separate-exhaust turbofan model, explicitly modelling: fan, core compressor, combustor, turbine (with power balance), core nozzle, and bypass nozzle — each with correct choked/unchoked treatment.

FR-06 The system shall implement a steady-state turboprop model, explicitly modelling: gas generator, power turbine, gearbox efficiency, propeller efficiency, propeller thrust (with static-condition momentum theory treatment), residual jet thrust, equivalent thrust, and ESHP.

FR-07 Every engine model shall perform validity checks before execution: altitude range, Mach range, throttle range, pressure ratio validity, efficiency bounds, and check for negative/NaN/complex results in intermediate calculations (see Section 18).

FR-08 The system shall execute engine models at a single operating point (single-point analysis), over an altitude vector (altitude sweep), or over a 3D grid of altitude × Mach × throttle (performance map generation).

FR-09 The system shall store simulation results in Performance_Data with Data_Type = 'Simulated', recording Engine_ID, Altitude, Mach, Throttle, Thrust_N, FuelFlow_kgs, TSFC_kgNs, Eta_thermal, Eta_propulsive, Model_Version, and timestamp.

FR-10 The system shall generate ML training data by executing the validated physics model over the approved 3D grid (Section 20), with invalid-condition filtering, and store it separately from Reference and operational Simulated data.

FR-11 The system shall train engine-architecture-specific ML surrogate models (architecture groups: turbojet, turbofan, turboprop) using at least four candidate algorithms: Polynomial Regression, Random Forest, Gaussian Process Regression, and ANN.

FR-12 The system shall evaluate each trained ML model on a held-out test set of physics-model data and report R^2, RMSE, MAE, MAPE, and prediction time. Results shall note clearly that these metrics measure agreement with the physics model, not with real engine measurements.

FR-13 The system shall implement a mission-based engine selection algorithm that: (a) applies hard constraints first, excluding non-feasible engines entirely; (b) applies weighted scoring to feasible engines only; (c) handles the zero-feasible-engine case by returning an explanatory message.

FR-14 The system shall provide at least five pre-defined mission weight presets (Fighter, Strike, Commercial, Transport, Trainer) and a custom weight entry mode that enforces $\sum w_i = 1.0$ before running the algorithm.

FR-15 The GUI shall display, for every simulation result: numeric values with units, model type (Physics/ML), data confidence level, and appropriate warnings for estimated engine data, out-of-range conditions, and ML unavailability.

FR-16 For the Kaveri engine, every GUI output shall display: "⚠ ESTIMATED / ACADEMIC MODEL — NOT MANUFACTURER-CONFIRMED PERFORMANCE".

FR-17 The system shall log every Digital Twin execution (L4 and above) to the Simulation_Log table with engine ID, model level, input conditions, outputs (JSON), timestamp, and model version.

5. Non-Functional Requirements

Table 5.1 Non-Functional Requirements

ID	Category	Requirement	Verification Method
NFR-01	Performance	Single-point physics model execution shall complete in <5 seconds per engine per operating point on a mid-range laptop (Intel i5 or AMD Ryzen 5, 8 GB RAM). ML surrogate prediction shall complete in <1 ms per point.	Timed test using MATLAB `tic/toc`
NFR-02	Accuracy	The ISA model shall reproduce ICAO Doc 7488/3 reference values within ±0.1%. Engine model tolerances are engine-specific and confidence-dependent (see Section 2 OBJ-03/04/05).	Unit test assertions
NFR-03	Maintainability	All MATLAB functions shall be documented with in-code comments, input/output argument descriptions, unit specifications, and valid-range annotations. Functions shall be no longer than 200 lines.	Code review
NFR-04	Portability	The database shall be a single SQLite `.db` file portable across operating systems. All paths shall use MATLAB `fullfile()` for OS-agnostic path construction.	Test on Windows and Linux

NFR-05	Reproducibility	ML model training shall use a fixed random seed (`rng(42,'twister')`). All parameter values shall be traceable to a source entry in Parameter_Provenance. Results shall be bit-reproducible on the same MATLAB version.	Re-run test
NFR-06	Data Integrity	All foreign key relationships shall be enforced by SQLite with PRAGMA foreign_keys = ON. All INSERT statements shall use parameterised queries. No CASCADE DELETE shall be permitted without explicit supervisor approval.	Database constraint test
NFR-07	Safety / Honesty	Model outputs shall carry appropriate confidence and provenance labels. The system shall never present ML predictions as physics results or estimated parameters as manufacturer-confirmed values. All Kaveri outputs shall include the mandated disclaimer.	GUI inspection
NFR-08	Usability	The GUI shall be self-explanatory for a third-year B.Tech student with no prior exposure to the project. All inputs shall display valid ranges; all outputs shall display units.	Usability walkthrough with uninstructed user
NFR-09	Version Control	The GitHub repository shall have a minimum of 20 commits, a structured directory tree, and a README describing installation, database setup, and usage.	Repository audit

6. Engine Categories

Gas turbine engines for aviation are classified by their thermodynamic cycle and the method by which thrust or shaft power is produced. Phase 1 of this project implements three architectures; Phase 2 and 3 define the roadmap for future expansion.

Table 6.1 Gas Turbine Engine Architecture Classification

Type	Phase	Propulsive Method	Typical BPR	Applications	Phase 1 Engines
Turbojet	Phase 1	Hot core exhaust jet only; no bypass or propeller	0 (single stream)	Supersonic fighters, early jet aircraft	J79, Atar 9C, R-25
Turbofan — Low-BPR Military	Phase 1	Core jet + small bypass jet; afterburner possible	0.15–0.50	Modern fighter aircraft, attack aircraft	F100-PW-229, M88-2, Kaveri

Turbofan — High-BPR Commercial	Phase 1	Large fan jet (dominant) + small core jet	4.5–10+	Commercial airliners, military transport	CFM56-5B4, CF6-80C2
Turboprop	Phase 1	Shaft power to propeller via gearbox; small residual jet	N/A	Regional airliners, military patrol, trainers	PT6A-67D, PW127G, AE 2100D3
Turboshaft	Phase 2	Shaft power only; no residual jet contribution	N/A	Helicopters, APU, ship propulsion	Planned Phase 2
Ramjet / Scramjet	Phase 3	Ram compression; no rotating machinery	N/A	High-speed missiles, hypersonic	Planned Phase 3
Combined cycle	Phase 3	Variable-cycle hybrid	Variable	Next-generation vehicles	Planned Phase 3

Turboprop vs Turboshaft Distinction (v1.1 Correction): A turboprop extracts shaft power from a free or coupled power turbine and delivers it through a reduction gearbox to a propeller. The engine is designed to produce net forward thrust, with residual jet thrust from the exhaust contributing typically 5–15% of total thrust equivalent. A turboshaft is configured to deliver shaft power with essentially no net jet thrust from the exhaust (used in helicopters where the rotor is the sole propulsive device). These are not interchangeable in Phase 1. The Klimov TV3-117, included in v1.0 as a "turboprop representative," is correctly classified as a turboshaft helicopter engine and has been replaced in v1.1 with the Pratt & Whitney Canada PW127G, a genuine turboprop with full EASA type certification data.

7. Engine Selection Criteria

Eleven representative engines are selected for Phase 1. The selection is guided by six criteria that balance technical scope, data availability, and contextual relevance. These criteria apply across all three architecture groups.

Table 7.1 Engine Selection Criteria

Rank	Criterion	Rationale	Application

1	Public-domain data availability	The engine must have sufficient publicly documented parameters (OPR, rated thrust or shaft power, dry weight, air mass flow, BPR, typical TIT range) to support cycle analysis without fabrication. This is the highest-priority filter — no engine with Unavailable data for core parameters shall be modelled without explicit disclaimers.	All engines
2	Architecture representativeness	Each of the three Phase 1 architecture groups (turbojet, turbofan, turboprop) must include at least one Low-BPR, one High-BPR, and one "extreme" design (e.g., very low OPR vs very high OPR). This ensures the models are exercised across their full parameterisation range.	All engines
3	IAF / DRDO operational relevance	Preference is given to engines that power aircraft in IAF service, engines under development by DRDO/GTRE, or engines that compete directly with IAF platforms. This ensures the engine selection tool has practical relevance for the intended audience.	Military engines
4	Parameter source confidence	At least two engines per architecture group must have High-confidence data (government type certificate, FAA/EASA TCDS, or extensive academic literature with cross-referenced values). At least one High-confidence engine per group is required to anchor validation.	Mandatory
5	Thermodynamic diversity	Engines are selected to provide a wide spread of OPR (5.8–32), TIT range, BPR (0–5.4), and rated power (895 kW to ~257 kN). This diversity ensures the database covers the full performance envelope relevant to comparative analysis.	Ensemble selection
6	Academic tractability	For engines with limited data (Kaveri, R-25), cycle analysis must still be possible using estimated parameters with clearly stated assumptions. The model must not require proprietary data to execute, even if its predictions are acknowledged as approximate.	Low-confidence engines

8. Engine Manufacturers

The 11 Phase 1 engines are produced by six manufacturers across five countries. Manufacturer characteristics determine public data availability and regulatory documentation.

Table 8.1 Engine Manufacturer Summary

Manufacturer	Country	Phase 1 Engines	Primary Regulatory Authority	Data Availability Notes

General Electric Aviation	USA	J79-GE-17, CF6-80C2B1	FAA (TCDS issued for civil variants)	Extensive NASA/NACA reports for J79; FAA TCDS for CF6 family. High confidence.
Safran Aircraft Engines (SNECMA)	France	Atar 9C, M88-2, CFM56-5B4 (joint)	DGAC France / EASA	Atar 9C: historical SNECMA publications, High confidence. M88-2: DGA press data + AIAA, Medium confidence. CFM56-5B4: EASA TCDS E.042, High confidence.
Pratt & Whitney / P&WC	USA / Canada	F100-PW-229, PT6A-67D, PW127G	FAA / Transport Canada / EASA	F100-PW-229: Jane's + AIAA, Medium. PT6A-67D: FAA TCDS E5EA, High. PW127G: EASA TCDS E.057, High.
Rolls-Royce	UK	AE 2100D3	EASA (TCDS E.003)	Product specification + EASA TCDS + open C-130J literature. High confidence.
Saturn/Tumansky (Russia)	Russia	R-25-300	Russian AR IAC (CSAR)	Basic parameters from declassified US intelligence; Jane's; academic papers. Medium confidence. No Western TC exists.
GTRE / DRDO (India)	India	GTX-35VS Kaveri	None (prototype/development)	DRDO press releases; parliamentary reports; IISc/NAL academic papers. Low confidence. No TCDS exists. All values estimated/assumed.

9. Engine Selection — Representative Engines

Data Classification Policy: All values are classified per the six-level source hierarchy: **MC = Manufacturer-Confirmed** | **TC = Type Certificate / Government Source** | Acad = Academic Literature | Calc = Calculated from known parameters | **Est = Estimated (engineering judgement)** | **Assum = Assumed (order-of-magnitude only)**. Confidence level: High (MC/TC sources), Medium (Acad/Calc), Low (Est/Assum). Every value entered into the database must have a corresponding Parameter_Provenance entry.

9.1 Turbojet Engines (3 Engines)

Turbojet engines for Phase 1 are selected to span a wide OPR range (5.8–8.8) and to represent the Soviet, French, and American engineering traditions present in IAF history.

Table 9.1 Phase 1 Turbojet Engine Specifications with Source Classification

Parameter	J79-GE-17A (USA)	SNECMA Atar 9C (France)	Klimov R-25-300 (Russia)
Application	F-4 Phantom II, F-104	Mirage III, Mirage 5, IV	MiG-21bis (IAF service)
Configuration	Single-spool, 17-stage axial, AB	Single-spool, 9-stage axial, AB	Single-spool, axial, AB
OPR	13.5 [Acad — Mattingly 2006, Table A.7; cross-referenced NASA TM-X-1669]	5.8 [Acad — multiple AIAA papers; Jane's 1972]	~8.8 [Est — derived from declassified DTIC reports; cross-check against known performance; Medium confidence]
Rated Thrust (dry, N)	52 489 N [TC — Jane's; DTIC ADA series, consistent with USAF operational data]	41 800 N [Acad — Jane's 1972; ONERA publications; consistent across multiple sources]	40 200 N dry; 69 600 N with afterburner [Est — DTIC translation; uncertainty ±15%]
Air Mass Flow (kg/s)	~77 kg/s [Acad — NASA TM-X-1669; Mattingly A.7]	~60 kg/s [Acad — Jane's; estimated from known thrust and TSFC]	~72 kg/s [Est — derived from thrust/OPR; Low-Medium confidence]
TSFC (dry, kg/N/hr)	~0.0230 kg/(N·hr) [Acad — NASA reports; Mattingly]	~0.0270 kg/(N·hr) [Acad — Jane's; estimated]	~0.0240 kg/(N·hr) dry [Est — Low-Medium confidence; ±20%]
Dry Weight (kg)	1 749 kg [TC — Jane's; consistent across US Air Force MDS documents]	1 438 kg [Acad — Jane's 1972]	~1 218 kg [Est — Soviet documentation translated by DTIC; Medium confidence]
Compressor stages	17 axial [MC — GE documentation]	9 axial [Acad — Jane's]	~6 axial [Est — from open sources]
Afterburner	Yes [MC]	Yes [MC]	Yes [Acad]
Overall Data Confidence	**HIGH**	**HIGH**	**MEDIUM**
Validation tolerance	±10%	±10%	±20%

R-25-300 Note: Basic performance data for the R-25-300 is derived from DTIC-translated Soviet documents and cross-referenced against MiG-21bis aircraft performance data. The engine was not exported with technical documentation accessible to Western researchers, and no EASA/FAA type certificate exists. All thermodynamic cycle parameters are marked Medium-to-Low confidence. The stated dry thrust of 40 200 N (dry) and 69 600 N (with afterburner) are consistent with Jane's reporting; however the specific

values used for cycle analysis (TIT, compressor efficiency, etc.) must be treated as Estimated with ±20% tolerance in validation.

9.2 Turbofan Engines (5 Engines)

Five turbofan engines are selected to cover BPR from 0.16 (Kaveri, very low) to 5.4 (CFM56-5B4, high-bypass commercial). Both low-BPR military and high-BPR commercial variants are modelled using the same separate-exhaust framework with different engine parameters.

Table 9.2 Phase 1 Turbofan Engine Specifications — Part A: High-Confidence Engines

Parameter	CFM56-5B4 (France/USA)	GE CF6-80C2B1 (USA)
Application	Airbus A320-200	Boeing 767-300, A300-600
BPR	5.4 [TC — EASA TCDS E.042]	5.15 [TC — FAA TCDS E6NE]
OPR	27.0 [TC — EASA TCDS E.042]	31.5 [TC — FAA TCDS E6NE]
Fan PR	~1.735 [Acad — estimated from OPR/OPR_core split]	~1.65 [Acad — estimated]
Rated Thrust (sea-level, N)	120 100 N [TC — EASA TCDS E.042, Takeoff rating]	257 300 N [TC — FAA TCDS E6NE, rated takeoff]
Air Mass Flow (kg/s)	~388 kg/s total; ~61 kg/s core [Acad — GE/Safran press; AIAA papers]	~787 kg/s total [Acad — GE Aviation documentation; NASA CR]
TSFC cruise (kg/N·s)	~0.0155 at 11 000 m, M0.82 [Acad — EASA TCDS; consistent with GE press]	~0.0158 at 11 000 m [Acad — FAA TCDS; GE Aviation]
Dry Weight (kg)	2 381 kg [TC — EASA TCDS E.042]	4 181 kg [TC — FAA TCDS E6NE]
Overall Data Confidence	**HIGH**	**HIGH**
Validation tolerance	±10%	±10%

Table 9.3 Phase 1 Turbofan Engine Specifications — Part B: Medium and Low Confidence

Parameter	F100-PW-229 (USA)	SNECMA M88-2 (France)	GTX-35VS Kaveri (India)
Application	F-15E, F-16C/D Block 40+	Dassault Rafale (IAF)	LCA Tejas Mk2 (proposed)
BPR	0.36 [Acad — Jane's; AIAA papers; consistent]	0.30 [Acad — DGA press; Safran; Jane's]	~0.16 [Est — DRDO press releases; academic papers. Low confidence. Some sources suggest 0.16–0.25]

OPR	32.0 [Acad — Jane's; P&W press; AIAA]	24.5 [Acad — Safran press; DGA; AIAA]	~21.0 [Est — DRDO/GTRE press releases; academic estimates. Low confidence. Unavailable from official type documentation — no TCDS]
Fan PR	~3.0 [Acad — estimated from OPR; AIAA N+2 study]	~3.2 [Acad — estimated from Rafale performance data]	~3.5 [Assum — engineering order-of-magnitude assumption for low-BPR military turbofan. Low confidence.]
Rated Thrust dry (N)	79 400 N dry; 129 000 N with AB [Acad — Jane's; P&W press; consistent across multiple sources. Note: some sources cite 79 400 N, others 76 000 N — uncertainty ±5%]	50 040 N dry; 74 800 N with AB [Acad — DGA/Safran press releases. Note: AB thrust sometimes cited as 75 000 N — uncertainty ±3%]	Target: 81 000 N with AB [Est/Assum — DRDO press conference data; parliamentary committee reports. Actual demonstrated thrust is NOT publicly available. This is a TARGET value, not achieved performance. Low confidence.]
Dry Weight (kg)	1 696 kg [Acad — Jane's; P&W press; consistent]	897 kg [Acad — Safran press; Jane's; AIAA]	~1 008 kg [Est — DRDO press; uncertainty ±15%. Low confidence]
Overall Data Confidence	**MEDIUM**	**MEDIUM**	**LOW**
Validation tolerance	±20%	±20%	±30% (model against targets, not measurements)

KAVERI (GTX-35VS) — CRITICAL DISCLAIMER: The Kaveri engine is under development by GTRE/DRDO and has not been type-certified or cleared for production. No manufacturer-confirmed thermodynamic cycle parameters are available in the public domain. All database values for this engine are marked **Confidence = 'Low', Source_Type = 'Estimated' or 'Assumed'**. Every simulation output for the Kaveri must display the mandatory warning: *"ESTIMATED / ACADEMIC MODEL — NOT MANUFACTURER-CONFIRMED PERFORMANCE. These results are generated from assumed cycle parameters and do NOT represent actual Kaveri engine performance."* The Kaveri is included in this project for its DRDO academic relevance, not as a reliable performance benchmark.

9.3 Turboprop Engines (3 Engines — v1.1 Revision)

Version 1.1 replaces the Klimov TV3-117VM (correctly classified as a turboshaft helicopter engine) with the Pratt & Whitney Canada PW127G, a genuine turboprop with full EASA type certification. Three authentic turboprop engines are now included.

v1.1 Correction — TV3-117VM Removed: The TV3-117VM was included in v1.0 as a "turboprop representative" for the IAF Mi-17 fleet. However, the TV3-117 family are turboshaft engines designed for helicopter applications, with exhaust configured to deliver no net thrust. They are not turboprops in any

standard engineering classification. The turboprop Phase 1 selection now consists of three genuine turboprops with reduction gearboxes and propeller drive: PT6A-67D, PW127G, and AE 2100D3. Turboshafts (Klimov TV3-117, Turbomeca Ardiden, etc.) are designated for Phase 2.

Table 9.4 Phase 1 Turboprop Engine Specifications with Source Classification

Parameter	P&WC PT6A-67D (Canada)	P&WC PW127G (Canada)	Rolls-Royce AE 2100D3 (UK)
Application	Beechcraft King Air 350, Pilatus PC-12	ATR 72-600, ATR 42-600	Lockheed C-130J Super Hercules (IAF)
Engine class	Free-turbine turboprop (reverse-flow)	Free-turbine turboprop (axial + centrifugal)	Turboprop (advanced, 6-blade Dowty propeller)
Max Shaft Power (kW)	895 kW (1200 shp) [TC — FAA TCDS E5EA rev.54]	2 051 kW (2 750 shp) [TC — EASA TCDS E.057 rev.5]	3 459 kW (4 637 shp) [TC — EASA TCDS E.003 rev.19]
OPR	~9.5 [Acad — Mattingly 2006; Walsh & Fletcher 2004; P&WC operator docs]	~16.0 [Acad — EASA E.057 background docs; AIAA regional turboprop studies]	~16.6 [Acad — EASA E.003; Rolls-Royce AE 2100 technical data; C-130J analyses]
Air Mass Flow (kg/s)	~5.5 kg/s [Acad — estimated from power/OPR; Mattingly PT6 family data]	~16 kg/s [Acad — estimated from power and OPR; AIAA]	~26 kg/s [Acad — estimated from power; C-130J analysis papers]
Dry Weight (kg)	213 kg [TC — FAA TCDS E5EA; P&WC documentation]	481 kg [TC — EASA TCDS E.057]	745 kg [TC — EASA TCDS E.003]
BSFC (kg/kW/hr)	~0.270 kg/kW/hr at max power [Acad — P&WC operator guide; Mattingly]	~0.258 kg/kW/hr at cruise [Acad — ATR performance data; EASA]	~0.223 kg/kW/hr [Acad — C-130J fuel data; Rolls-Royce AE 2100 documentation]
Propeller type	4-blade composite (Hartzell)	6-blade composite (Hamilton Sundstrand)	6-blade composite (Dowty R391)
Reduction gear ratio	~17:1 [Acad — P&WC technical documentation]	~19:1 [Acad — estimated; EASA E.057]	~17.8:1 [Acad — Rolls-Royce AE 2100 documentation]
Overall Data Confidence	**HIGH**	**HIGH**	**HIGH**
Validation tolerance	±10%	±10%	±15%

9.4 Thermodynamic Cycle Parameters for Database Population

The following parameters are required in the Engine_Thermo_Params table for each engine. Parameters marked *Unavailable* cannot be obtained from reliable public-domain sources and must not be fabricated. They are populated with NULL in the database and excluded from model execution for the corresponding

engine, or substituted with class-average Estimated values with explicit Low confidence tagging.

Table 9.5 Database Thermodynamic Parameters — Availability Matrix

Parameter	J79	Atar 9C	R-25	CFM56-5B4	CF6-80C2	F100	M88-2	Kaveri	PT6A-67D	PW127G	AE 2100D3
OPR	H	H	M	H	H	M	M	L	H	H	H
Fan PR (turbofan only)	—	—	—	M	M	M	M	L	—	—	—
BPR (turbofan only)	—	—	—	H	H	M	M	L	—	—	—
TIT / T_t4 (K)	M	M	L	M	M	L	L	L	M	M	M
Rated thrust / shaft power	H	H	M	H	H	M	M	L	H	H	H
Air mass flow	M	M	L	M	M	L	L	L	M	M	M
Dry weight	H	H	M	H	H	M	M	L	H	H	H
Compressor efficiency η_c	L	L	L	M	M	L	L	L	M	M	M
Turbine efficiency η_t	L	L	L	M	M	L	L	L	M	M	M
Propeller efficiency (turboprop)	—	—	—	—	—	—	—	—	M	M	M

H = High confidence (TC/MC source); M = Medium confidence (Academic/Calculated); L = Low confidence (Estimated/Assumed); — = Not applicable to this architecture

10. Database Architecture

10.1 Technology Selection — SQLite

Selected Technology: SQLite 3.x accessed via the MATLAB Database Toolbox JDBC driver (`org.sqlite.JDBC`). This choice is optimal for the academic constraints of this project: zero server infrastructure, single portable file, full SQL-92 compliance with JOINs, subqueries, views, and transactions, and referential integrity via `PRAGMA foreign_keys = ON`. The projected database size is under 100 MB — well within SQLite's practical performance envelope.

10.2 Design Principles (v1.1 Revised)

- **Third Normal Form (3NF):** All tables are normalised to 3NF. Manufacturer data is stored in a separate `Manufacturers` table; engine type in `Engine_Types`; every non-key attribute depends on the full key and only the key. No transitive dependencies exist between non-key columns.
- **Per-Parameter Source Traceability (v1.1 Addition):** A new `Parameter_Provenance` table provides a dedicated record for every externally sourced numerical value in `Engine_Thermo_Params`. This replaces the v1.0 approach of one reference row per engine, which could not trace individual parameters to their specific sources. Every populated thermo parameter must have a corresponding Parameter_Provenance row before the engine can be used in model execution.
- **Referential Integrity:** All foreign keys are declared in the schema. `PRAGMA foreign_keys = ON` is set at every connection. Cascade behaviour is restricted: no CASCADE DELETE without supervisor approval; RESTRICT is the default for all child tables.
- **Data Type Separation:** The `Performance_Data` table uses a mandatory `Data_Type` column (values: 'Reference', 'Simulated', 'ML_Predicted') to prevent provenance mixing. Reference data (from published sources) is never overwritten by simulation results. ML data is generated in a separate batch and tagged accordingly.
- **Parameterised Queries:** All MATLAB SQL queries use the `?` placeholder with cell array binding to prevent SQL injection and ensure correct type handling.
- **Transaction Handling:** Multi-row INSERT operations (e.g., loading performance map data) are wrapped in explicit `BEGIN / COMMIT` transactions. Failed transactions are rolled back atomically.

10.3 MATLAB Database Toolbox Integration

```
% Connection pattern — always use parameterised queries
conn = database('engine_performance.db', '', '', 'org.sqlite.JDBC', ...
    ['jdbc:sqlite:' fullfile(projectRoot, 'database', 'engine_performance.db')]);
execute(conn, 'PRAGMA foreign_keys = ON');

% Parameterised SELECT
results = fetch(conn, 'SELECT * FROM Engines WHERE Type_ID = ?', {typeID});

% Parameterised INSERT with explicit column list
insert_sql = ['INSERT INTO Performance_Data (Engine_ID, Altitude_m, Mach, Throttle, '...
    'Thrust_N, FuelFlow_kgs, TSFC_kgNs, Eta_thermal, Eta_propulsive, Data_Type, '...
    'Model_Version, Timestamp_UTC) VALUES (?,?,?,?,?,?,?,?,?,?,?,?)'];
execute(conn, insert_sql, {eng_id, alt, mach, thr, F, ff, tsfc, eta_th, eta_pr, ...
    'Simulated', '1.0', datestr(now,'yyyy-mm-dd HH:MM:SS')});

% Always close connection in finally block equivalent
close(conn);
```

11. Database Schema

The complete Phase 1 schema comprises 11 tables (10 from v1.0 plus the new `Parameter_Provenance` table). All tables use `INTEGER PRIMARY KEY AUTOINCREMENT` as surrogate keys. `PRAGMA foreign_keys = ON` must be set before any DML operation.

11.1 Manufacturers

```
CREATE TABLE Manufacturers (
    Manufacturer_ID  INTEGER PRIMARY KEY AUTOINCREMENT,
    Name             TEXT NOT NULL UNIQUE,
    Country          TEXT NOT NULL,
    Website          TEXT,
    Notes            TEXT
);
```

11.2 Engine_Types

```
CREATE TABLE Engine_Types (
    Type_ID     INTEGER PRIMARY KEY AUTOINCREMENT,
     Type_Name    TEXT NOT NULL UNIQUE    -- e.g. 'Turbojet', 'Turbofan_LowBPR', 'Turbofan_HighBPR',
'Turboprop', 'Turboshaft'
    Phase       INTEGER NOT NULL DEFAULT 1,  -- 1=Phase1, 2=Phase2, etc.
    Description TEXT
);
```

11.3 Engines

```
CREATE TABLE Engines (
    Engine_ID       INTEGER PRIMARY KEY AUTOINCREMENT,
    Designation     TEXT NOT NULL UNIQUE,    -- e.g. 'J79-GE-17A'
    Manufacturer_ID INTEGER NOT NULL,
    Type_ID         INTEGER NOT NULL,
    First_Run_Year  INTEGER,
    Status          TEXT NOT NULL DEFAULT 'Active',   -- 'Active','Development','Retired'
    Afterburner     INTEGER NOT NULL DEFAULT 0,       -- 0=No, 1=Yes (SQLite BOOLEAN)
    Notes           TEXT,
    FOREIGN KEY (Manufacturer_ID) REFERENCES Manufacturers(Manufacturer_ID) ON DELETE RESTRICT,
    FOREIGN KEY (Type_ID) REFERENCES Engine_Types(Type_ID) ON DELETE RESTRICT
);
```

11.4 Engine_Thermo_Params

```
CREATE TABLE Engine_Thermo_Params (
    Param_ID           INTEGER PRIMARY KEY AUTOINCREMENT,
    Engine_ID          INTEGER NOT NULL UNIQUE,   -- one thermo-params row per engine
    OPR                REAL,                      -- Overall pressure ratio [-]
     Fan_PR                REAL,                          -- Fan pressure ratio (turbofan only; NULL
otherwise)
    BPR                REAL,                      -- Bypass ratio (turbofan only; NULL otherwise)
    T_t4_K             REAL,                      -- Turbine inlet total temperature [K]
    MassFlow_kgs       REAL,                      -- Corrected air mass flow at design point [kg/s]
    Eta_compressor     REAL CHECK(Eta_compressor IS NULL OR (Eta_compressor > 0 AND Eta_compressor <=
1)),
    Eta_fan            REAL CHECK(Eta_fan IS NULL OR (Eta_fan > 0 AND Eta_fan <= 1)),
     Eta_combustor       REAL CHECK(Eta_combustor IS NULL OR (Eta_combustor > 0 AND Eta_combustor <=
1)),
    Eta_turbine        REAL CHECK(Eta_turbine IS NULL OR (Eta_turbine > 0 AND Eta_turbine <= 1)),
    Eta_nozzle_core    REAL CHECK(Eta_nozzle_core IS NULL OR (Eta_nozzle_core > 0 AND Eta_nozzle_core
<= 1)),
      Eta_nozzle_bypass   REAL  CHECK(Eta_nozzle_bypass  IS  NULL  OR  (Eta_nozzle_bypass  IS  NULL  OR
(Eta_nozzle_bypass > 0 AND Eta_nozzle_bypass <= 1))),
    Eta_mechanical     REAL CHECK(Eta_mechanical IS NULL OR (Eta_mechanical > 0 AND Eta_mechanical <=
1)),
    Eta_gearbox        REAL CHECK(Eta_gearbox IS NULL OR (Eta_gearbox > 0 AND Eta_gearbox <= 1)),  --
turboprop only
     Eta_propeller       REAL CHECK(Eta_propeller IS NULL OR (Eta_propeller > 0 AND Eta_propeller <=
1)), -- turboprop only
    Prop_Diameter_m    REAL,                      -- Propeller disc diameter [m] (turboprop only)
    Burner_PR_loss     REAL NOT NULL DEFAULT 0.05, -- Fractional total pressure loss in combustor [-]
    Inlet_PR_recovery  REAL NOT NULL DEFAULT 0.97, -- Subsonic inlet total pressure recovery [-]
     Rated_Thrust_N      REAL,                         -- Rated dry thrust [N] (jet engines); NULL for
turboprop
    Rated_Power_kW     REAL,                       -- Rated shaft power [kW] (turboprop); NULL for jet
engines
    DryWeight_kg       REAL,
            Overall_Confidence   TEXT   NOT   NULL   DEFAULT   'Low'   CHECK(Overall_Confidence   IN
('High','Medium','Low')),
    FOREIGN KEY (Engine_ID) REFERENCES Engines(Engine_ID) ON DELETE RESTRICT
);
```

11.5 Parameter_Provenance (New in v1.1)

This table provides per-parameter source traceability. Every numerical value in `Engine_Thermo_Params` that comes from an external source must have a corresponding row here. This enables the GUI to display the source and confidence of any displayed value on demand.

```
CREATE TABLE Parameter_Provenance (
    Provenance_ID    INTEGER PRIMARY KEY AUTOINCREMENT,
    Engine_ID        INTEGER NOT NULL,
    Parameter_Name   TEXT NOT NULL,   -- Column name in Engine_Thermo_Params, e.g. 'OPR', 'T_t4_K'
    Parameter_Value  REAL,            -- The actual value stored (for audit cross-check)
    Source_Type      TEXT NOT NULL CHECK(Source_Type IN (
                         'Manufacturer_Confirmed','Type_Certificate','Academic_Literature',
                         'Calculated','Estimated','Assumed','Unavailable')),
    Confidence_Level TEXT NOT NULL CHECK(Confidence_Level IN ('High','Medium','Low')),
    Source_Reference TEXT NOT NULL,   -- Full citation: author/document/year/section, e.g. 'EASA TCDS
E.042 Rev.17, Table 1'
    Notes            TEXT,            -- Any caveats, e.g. 'Uncertainty ±15%'
    Date_Added       TEXT NOT NULL DEFAULT (date('now')),
    FOREIGN KEY (Engine_ID) REFERENCES Engines(Engine_ID) ON DELETE RESTRICT,
    UNIQUE (Engine_ID, Parameter_Name)
);
```

11.6 Aircraft and Engine_Aircraft_Map

```
CREATE TABLE Aircraft (
    Aircraft_ID  INTEGER PRIMARY KEY AUTOINCREMENT,
    Designation  TEXT NOT NULL UNIQUE,  -- e.g. 'F-4C Phantom II'
    Type         TEXT,                  -- 'Fighter', 'Bomber', 'Transport', 'Trainer', 'Airliner'
    Country      TEXT,
    IAF_Service  INTEGER DEFAULT 0      -- 1 if in IAF service
);

CREATE TABLE Engine_Aircraft_Map (
    Map_ID      INTEGER PRIMARY KEY AUTOINCREMENT,
    Engine_ID   INTEGER NOT NULL,
    Aircraft_ID INTEGER NOT NULL,
    Notes       TEXT,
    FOREIGN KEY (Engine_ID) REFERENCES Engines(Engine_ID) ON DELETE RESTRICT,
    FOREIGN KEY (Aircraft_ID) REFERENCES Aircraft(Aircraft_ID) ON DELETE RESTRICT,
    UNIQUE(Engine_ID, Aircraft_ID)
);
```

11.7 Performance_Data

```
CREATE TABLE Performance_Data (
    Data_ID         INTEGER PRIMARY KEY AUTOINCREMENT,
    Engine_ID       INTEGER NOT NULL,
    Data_Type       TEXT NOT NULL CHECK(Data_Type IN ('Reference','Simulated','ML_Predicted')),
    Altitude_m      REAL NOT NULL,
    Mach            REAL NOT NULL,
    Throttle        REAL NOT NULL CHECK(Throttle >= 0.0 AND Throttle <= 1.0),
    Afterburner     INTEGER NOT NULL DEFAULT 0,
    Thrust_N        REAL,             -- Net thrust [N] or NULL for turboprop
    ShaftPower_kW   REAL,             -- Shaft power [kW] or NULL for jet engines
    FuelFlow_kgs    REAL NOT NULL,
    TSFC_kgNs       REAL,             -- Thrust TSFC [kg/(N·s)] or NULL for turboprop
    BSFC_kgkWh      REAL,             -- Shaft BSFC [kg/(kW·h)] or NULL for jet engines
    Eta_thermal     REAL,
    Eta_propulsive  REAL,
    Source_Ref      TEXT,             -- If Data_Type='Reference', citation; else NULL
    Model_Version   TEXT,             -- e.g. '1.0' for physics model version
    ML_Model_Type   TEXT,             -- If Data_Type='ML_Predicted', model name, e.g. 'RF'
    Timestamp_UTC   TEXT NOT NULL DEFAULT (datetime('now')),
    FOREIGN KEY (Engine_ID) REFERENCES Engines(Engine_ID) ON DELETE RESTRICT
);
```

11.8 Mission_Profiles

```
CREATE TABLE Mission_Profiles (
    Mission_ID     INTEGER PRIMARY KEY AUTOINCREMENT,
    Mission_Name   TEXT NOT NULL UNIQUE,
    Altitude_m     REAL NOT NULL,
    Mach           REAL NOT NULL CHECK(Mach >= 0),
    Min_Thrust_N   REAL,              -- Hard constraint: minimum net thrust required [N]
    Max_TSFC_kgNs  REAL,              -- Hard constraint: maximum TSFC [kg/(N·s)]
    W_Thrust       REAL NOT NULL CHECK(W_Thrust >= 0 AND W_Thrust <= 1),
    W_Efficiency   REAL NOT NULL CHECK(W_Efficiency >= 0 AND W_Efficiency <= 1),
    W_Fuel         REAL NOT NULL CHECK(W_Fuel >= 0 AND W_Fuel <= 1),
    W_Altitude     REAL NOT NULL CHECK(W_Altitude >= 0 AND W_Altitude <= 1),
    W_TWR          REAL NOT NULL CHECK(W_TWR >= 0 AND W_TWR <= 1),
    Notes          TEXT,
    CHECK(ABS(W_Thrust + W_Efficiency + W_Fuel + W_Altitude + W_TWR - 1.0) < 0.001)
);
```

11.9 References

```
CREATE TABLE References (
    Ref_ID       INTEGER PRIMARY KEY AUTOINCREMENT,
    Short_Tag    TEXT NOT NULL UNIQUE,    -- e.g. 'Mattingly2006', 'EASA_E042_R17'
    Authors      TEXT,
    Title        TEXT NOT NULL,
    Publisher    TEXT,
    Year         INTEGER,
    Source_Type  TEXT CHECK(Source_Type IN (
                     'Type_Certificate','Government_Report','Academic_Journal',
                     'Conference_Paper','Textbook','Press_Release','Web_Database')),
    URL_DOI      TEXT,
    Notes        TEXT
);
```

11.10 Simulation_Log

```
CREATE TABLE Simulation_Log (
    Log_ID          INTEGER PRIMARY KEY AUTOINCREMENT,
    Engine_ID       INTEGER NOT NULL,
    DT_Level        TEXT NOT NULL CHECK(DT_Level IN ('L1','L2','L3','L4','L5')),
    Model_Type      TEXT NOT NULL CHECK(Model_Type IN ('Physics','ML')),
    ML_Model_Name   TEXT,             -- e.g. 'RF','GPR','ANN','Poly'; NULL if Physics
    Altitude_m      REAL,
    Mach            REAL,
    Throttle        REAL,
    Inputs_JSON     TEXT,             -- Full JSON of all inputs (for reproducibility)
    Outputs_JSON    TEXT,             -- Full JSON of all outputs
    Exec_Time_ms    REAL,
    Model_Version   TEXT,
    Timestamp_UTC   TEXT NOT NULL DEFAULT (datetime('now')),
    FOREIGN KEY (Engine_ID) REFERENCES Engines(Engine_ID) ON DELETE RESTRICT
);
```

12. Primary and Foreign Keys

Table 12.1 Primary Key and Foreign Key Summary

Table	Primary Key	Foreign Keys (column → parent table)	UNIQUE Constraints	Key Design Rationale
Manufacturers	Manufacturer_ID	—	Name	Natural key (Name) is too long for FK use; surrogate int key chosen. UNIQUE constraint on Name prevents duplicates.

Engine_Types	Type_ID	—	Type_Name	Small lookup table. Surrogate key allows renaming without cascading updates.
Engines	Engine_ID	Manufacturer_ID → Manufacturers; Type_ID → Engine_Types	Designation	Every engine references exactly one manufacturer and one type. UNIQUE on Designation ensures no two entries represent the same engine.
Engine_Thermo_Params	Param_ID	Engine_ID → Engines (RESTRICT)	Engine_ID	One-to-one with Engines. UNIQUE(Engine_ID) enforces this. Architecture-specific fields use NULL for inapplicable columns (e.g., Fan_PR = NULL for turbojets).
Parameter_Provenance	Provenance_ID	Engine_ID → Engines (RESTRICT)	(Engine_ID, Parameter_Name)	One row per engine-parameter pair. Composite UNIQUE prevents duplicate provenance records for the same parameter.
Aircraft	Aircraft_ID	—	Designation	Independent table; engines and aircraft are linked via the mapping table.
Engine_Aircraft_Map	Map_ID	Engine_ID → Engines; Aircraft_ID → Aircraft (both RESTRICT)	(Engine_ID, Aircraft_ID)	Many-to-many bridge table. Composite UNIQUE prevents the same pairing being inserted twice.
Performance_Data	Data_ID	Engine_ID → Engines (RESTRICT)	—	No UNIQUE constraint — multiple simulation runs can produce multiple rows at the same condition. Data_Type column separates Reference/Simulated/ML_Predicted provenance.
Mission_Profiles	Mission_ID	—	Mission_Name	No FK to engines — missions reference engine data at query time, not at storage time. CHECK constraint enforces weight sum = 1.0 ± 0.001.
References	Ref_ID	—	Short_Tag	Bibliography table. Referenced by Parameter_Provenance.Source_Reference (text citation, not FK, to allow flexible citation formats).

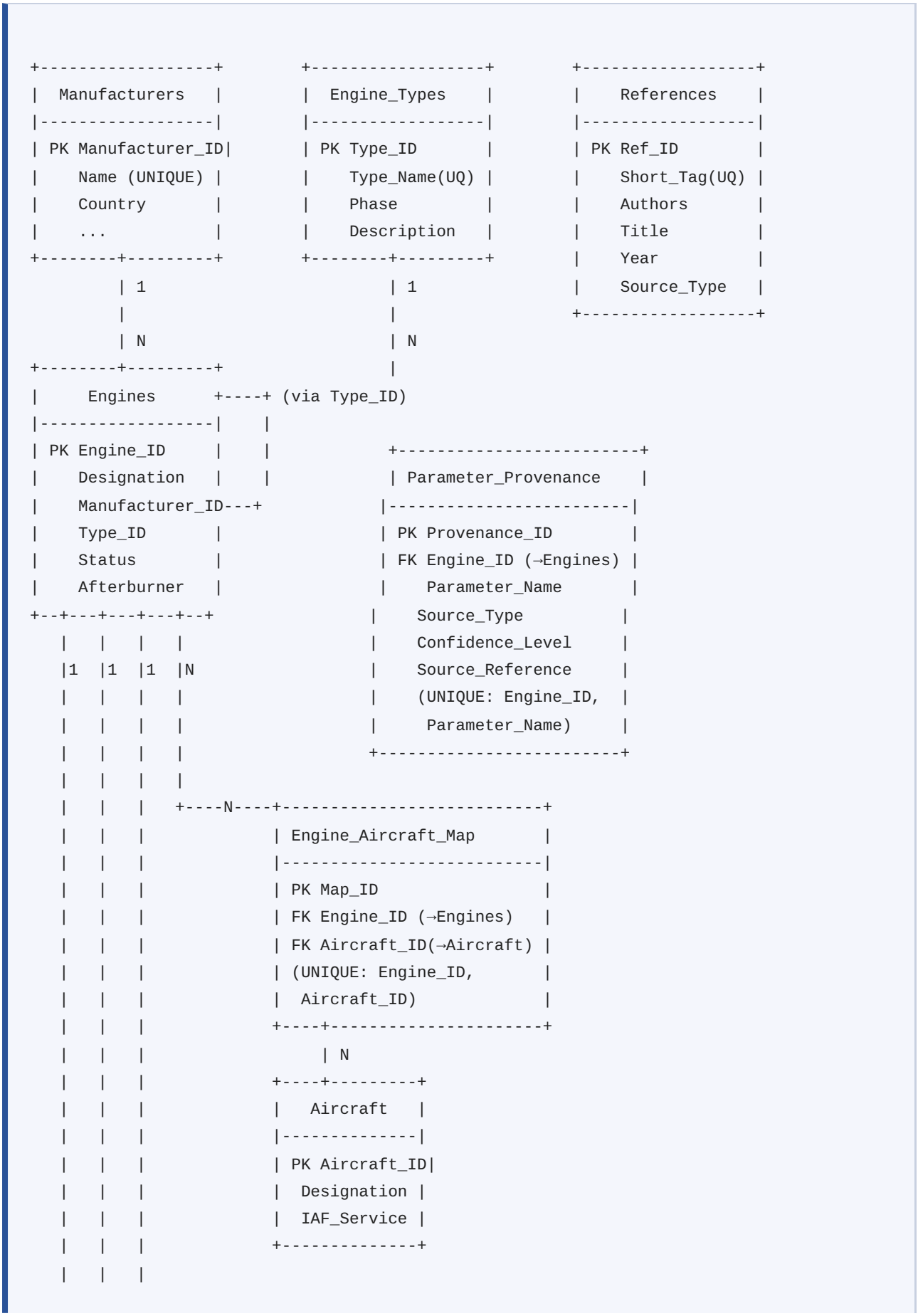
Simulation_Log	Log_ID	Engine_ID → Engines (RESTRICT)	—	Append-only audit table. No UPDATE or DELETE on existing log rows.

12.1 Cascading Behaviour Policy

All foreign keys use `ON DELETE RESTRICT` as the default. This means an engine record cannot be deleted while child records (thermo params, performance data, simulation logs) exist. This is intentional: data must be explicitly cleaned up before the parent engine record is removed, preventing accidental orphaning. No `ON DELETE CASCADE` is used in Phase 1 without explicit supervisor approval. `ON UPDATE CASCADE` is not needed since all PKs are surrogate integers that are never updated after insertion.

13. Entity-Relationship Diagram

The diagram below shows all 11 Phase 1 tables with their primary keys, foreign key relationships, and cardinality. Lines with single bars indicate a required (mandatory) relationship; crow's-foot notation indicates the "many" side. All relationships are 1:N or M:N (bridge table).


Manufacturers
PK Manufacturer_ID
Name (UNIQUE)
Country
...
Engine_Types
PK Type_ID
Type_Name(UQ)
Phase
Description
References
PK Ref_ID
Short_Tag(UQ)
Authors
Title
Year
Source_Type
Engines
(via Type_ID)
PK Engine_ID
Designation
Manufacturer_ID
Type_ID
Status
Afterburner
Parameter_Provenance
PK Provenance_ID
FK Engine_ID (→Engines)
Parameter_Name
Source_Type
Confidence_Level
Source_Reference
(UNIQUE: Engine_ID, Parameter_Name)
Engine_Aircraft_Map
PK Map_ID
FK Engine_ID (→Engines)
FK Aircraft_ID(→Aircraft)
(UNIQUE: Engine_ID, Aircraft_ID)
Aircraft
PK Aircraft_ID
Designation
IAF_Service

13.1 Key Relationships Summary

Table 13.1 Entity-Relationship Summary

Relationship	Cardinality	Notes
Manufacturers → Engines	1:N	One manufacturer produces many engines
Engine_Types → Engines	1:N	One type classifies many engines
Engines → Engine_Thermo_Params	1:1	UNIQUE(Engine_ID) enforces exactly one thermo-params row per engine
Engines → Parameter_Provenance	1:N	One engine has many parameter provenance rows (one per parameter)
Engines ↔ Aircraft (via Map)	M:N	One engine can power many aircraft; one aircraft can have multiple engine variants
Engines → Performance_Data	1:N	One engine has many performance data points (Reference + Simulated + ML)
Engines → Simulation_Log	1:N	One engine accumulates many simulation log entries over time
Mission_Profiles (standalone)	—	References engines at query time only; no stored FK relationship

14. Data Fields Summary

The master data dictionary below lists all performance-relevant fields, their units, valid ranges, applicable architectures, and whether a field requires a Parameter_Provenance entry.

Table 14.1 Master Data Field Dictionary

Field	Table	Unit	Type	Valid Range	Architecture	Provenance Required?
OPR	Engine_Thermo_Params	—	REAL	1.5 – 50	All	Yes
Fan_PR	Engine_Thermo_Params	—	REAL / NULL	1.1 – 5.0	Turbofan only	Yes (if not NULL)
BPR	Engine_Thermo_Params	—	REAL / NULL	0.10 – 15.0	Turbofan only	Yes
T_t4_K	Engine_Thermo_Params	K	REAL	900 – 2 200	All	Yes
MassFlow_kgs	Engine_Thermo_Params	kg/s	REAL	1 – 1500	All	Yes

Eta_compressor	Engine_Thermo_Params	—	REAL	0.70 – 0.95	All	Yes
Eta_fan	Engine_Thermo_Params	—	REAL / NULL	0.75 – 0.92	Turbofan only	Yes
Eta_combustor	Engine_Thermo_Params	—	REAL	0.95 – 0.999	All	Yes
Eta_turbine	Engine_Thermo_Params	—	REAL	0.80 – 0.95	All	Yes
Eta_nozzle_core	Engine_Thermo_Params	—	REAL	0.95 – 0.995	All	Yes
Eta_nozzle_bypass	Engine_Thermo_Params	—	REAL / NULL	0.95 – 0.995	Turbofan only	Yes
Eta_mechanical	Engine_Thermo_Params	—	REAL	0.96 – 0.999	All	Yes
Eta_gearbox	Engine_Thermo_Params	—	REAL / NULL	0.96 – 0.998	Turboprop only	Yes
Eta_propeller	Engine_Thermo_Params	—	REAL / NULL	0.75 – 0.92	Turboprop only	Yes
Prop_Diameter_m	Engine_Thermo_Params	m	REAL / NULL	0.5 – 5.5	Turboprop only	Yes
Burner_PR_loss	Engine_Thermo_Params	—	REAL	0.02 – 0.08	All	Yes
Rated_Thrust_N	Engine_Thermo_Params	N	REAL / NULL	>0	Jet engines	Yes
Rated_Power_kW	Engine_Thermo_Params	kW	REAL / NULL	>0	Turboprop only	Yes
DryWeight_kg	Engine_Thermo_Params	kg	REAL	50 – 10 000	All	Yes
Altitude_m	Performance_Data	m (geometric)	REAL	0 – 32 000	All	No (computed)
Mach	Performance_Data	—	REAL	0 – 3.0	Type-dependent	No
Throttle	Performance_Data	—	REAL	0.0 – 1.0	All	No
Thrust_N	Performance_Data	N	REAL / NULL	>0 or NULL	Jet engines	If Reference=Yes

TSFC_kgNs	Performance_Data	kg/(N·s)	REAL / NULL	>0	Jet engines	If Reference=Yes
ShaftPower_kW	Performance_Data	kW	REAL / NULL	>0	Turboprop only	If Reference=Yes
BSFC_kgkWh	Performance_Data	kg/(kW·h)	REAL / NULL	>0	Turboprop only	If Reference=Yes
Data_Type	Performance_Data	—	TEXT	'Reference','Simulated','ML_Predicted'	All	Mandatory

15. Data Sources

All data used in this project is sourced from public-domain documents. The five source categories are listed below in decreasing order of confidence. The full bibliography is maintained in the database References table; the entries below are illustrative of the key sources for each category.

Table 15.1 Primary Data Source Categories

Category	Confidence	Key Sources	Engines Covered
Type Certificates and Government Documents	High	EASA TCDS E.042 (CFM56-5B4); EASA E.057 (PW127G); EASA E.003 (AE 2100D3); FAA TCDS E5EA (PT6A-67D); FAA TCDS E6NE (CF6-80C2)	CFM56-5B4, CF6-80C2, PT6A-67D, PW127G, AE 2100D3
Academic Textbooks and NASA Reports	High–Medium	Mattingly, J.D. (2006) *Elements of Propulsion*; Walsh & Fletcher (2004) *Gas Turbine Performance*; Saravanamuttoo et al. (2017); NASA TM-X-1669; NASA CR series	J79, Atar 9C, PT6A-67D (cycle analysis defaults)
Jane's All the World's Aircraft (various editions)	Medium	Jane's 1972 (Atar 9C), Jane's 1985 (J79), Jane's 2010 (F100, M88-2, R-25). Jane's data is secondary and must be cross-referenced where possible.	All engines
AIAA Papers and Academic Journals	Medium	AIAA 2003-0304 (turbofan cycle analysis); AIAA 2008-4598 (military turbofan parametric); multiple ISABE and ASME Turbo Expo papers	Turbofan cycle parameters

DRDO/GTRE Press Releases and Parliamentary Reports	Low	MoD Annual Reports (2018–2024); DRDO Technology Focus issues; Parliamentary Standing Committee on Defence; IISc/NAL academic papers on Kaveri cycle analysis	Kaveri/GTX-35VS only

> **DTIC / Declassified Documents:** Some parameters for the R-25-300 and J79 are drawn from DTIC-declassified US government documents. These are treated as Medium confidence due to possible date-of-declassification uncertainty and limited cross-referencing opportunities. They are cited with full DTIC accession numbers in the References table.

16. Atmospheric Model (ISA)

16.1 Altitude Convention

The ISA model used throughout this project is formulated in terms of **geopotential altitude** H, which is the conventional reference altitude used in ICAO Doc 7488/3 and all standard atmosphere tables. The relationship between geometric altitude h and geopotential altitude H is:

$$H = \frac{R_E \cdot h}{R_E + h}$$

where $R_E = 6{,}356{,}766$ m is the standard Earth radius. For altitudes below 32 km, the difference between h and H is less than 0.16%, which is negligible for the present study. Throughout this document, the symbol H denotes geopotential altitude and all ISA equations are expressed in terms of H.

> **Assumption AS-ATM-01:** This project uses geopotential altitude throughout. Geometric altitude is accepted as input from the GUI and converted to geopotential altitude internally. The 0.16% approximation error is documented and accepted for academic comparative analysis.

16.2 ISA Equations by Layer

The ISA is divided into three layers relevant to the flight envelopes of the engines modelled in Phase 1.

Layer 1 — Troposphere (0 ≤ H ≤ 11,000 m)

Temperature lapse rate: $L = -6.5 \times 10^{-3}$ K/m

$$T(H) = T_0 + L \cdot H = 288.15 - 0.0065H \quad [\text{K}]$$

$$P(H) = P_0 \left(\frac{T(H)}{T_0} \right)^{g_0/(|L|R)} = 101{,}325 \left(\frac{T(H)}{288.15} \right)^{5.2561} \quad [\text{Pa}]$$

$$\rho(H) = \frac{P(H)}{R_{\text{air}}\, T(H)} \quad [\text{kg/m}^3]$$

where $g_0 = 9.80665$ m/s², $R_{\text{air}} = 287.058$ J/(kg·K), $P_0 = 101{,}325$ Pa, $T_0 = 288.15$ K.

Layer 2 — Lower Stratosphere (11,000 m < H ≤ 20,000 m)

Temperature is isothermal at the tropopause temperature $T_{11} = 216.65$ K:

$$T(H) = 216.65 \quad [\text{K}]$$

$$P(H) = P_{11} \exp\left(-\frac{g_0 (H - 11{,}000)}{R_{\text{air}}\, T_{11}} \right) \quad [\text{Pa}]$$

where $P_{11} = 22{,}632.1$ Pa (ISA pressure at $H = 11{,}000$ m).

Layer 3 — Upper Stratosphere (20,000 m < H ≤ 32,000 m)

Temperature lapse rate: $L = +1.0 \times 10^{-3}$ K/m (positive, temperature increases)

$$T(H) = 216.65 + 0.001 (H - 20{,}000) \quad [\text{K}]$$

$$P(H) = P_{20} \left(\frac{T(H)}{T_{20}} \right)^{-g_0/(L\, R_{\text{air}})} \quad [\text{Pa}]$$

where $P_{20} = 5{,}474.89$ Pa and $T_{20} = 216.65$ K are conditions at $H = 20{,}000$ m.

16.3 Derived Atmospheric Properties

Speed of sound:

$$a(H) = \sqrt{\gamma_{\text{air}}\, R_{\text{air}}\, T(H)} \quad [\text{m/s}] \quad \gamma_{\text{air}} = 1.4$$

Dynamic viscosity (Sutherland's law):

$$\mu(H) = \mu_{\text{ref}} \left(\frac{T(H)}{T_{\text{ref}}} \right)^{3/2} \frac{T_{\text{ref}} + S}{T(H) + S} \quad [\text{Pa s}]$$

where $\mu_{\text{ref}} = 1.716 \times 10^{-5}$ Pa·s, $T_{\text{ref}} = 273.15$ K, $S = 110.4$ K (Sutherland constant).

Kinematic viscosity: $\nu(H) = \mu(H) / \rho(H)$

16.4 Stagnation (Total) Conditions

For a flow at Mach number M and static temperature T and static pressure P:

$$T_t = T\left(1 + \frac{\gamma_{\text{air}} - 1}{2} M^2\right) \quad [\text{K}]$$

$$P_t = P\left(1 + \frac{\gamma_{\text{air}} - 1}{2} M^2\right)^{\gamma_{\text{air}}/(\gamma_{\text{air}}-1)} \quad [\text{Pa}]$$

For air, $\gamma_{\text{air}} = 1.4$ and the exponent $\gamma/(\gamma - 1) = 3.5$. These relations assume isentropic, calorically perfect gas — a standard assumption justified for $M \leq 3$ and $T \leq 800$ K.

> **Model Scope:** All engine thermodynamic models in §17 are supplied stagnation conditions at the engine inlet (Station 0/2) computed from the ISA layer equations above plus the Mach number input. The models then work in total properties throughout.

16.5 ISA Validation Table

The table below compares model outputs against ICAO Doc 7488/3 standard values. All values are at ISA standard conditions (no temperature offset).

Table 16.1 ISA model validation against ICAO Doc 7488/3 reference values

Alt. H (m)	T [K] Model	T [K] ICAO	P [Pa] Model	P [Pa] ICAO	ρ [kg/m³] Model	ρ [kg/m³] ICAO	a [m/s] Model	a [m/s] ICAO
0	288.150	288.150	101,325	101,325	1.2250	1.2250	340.29	340.29
2,000	275.150	275.154	79,495	79,501	1.0066	1.0066	332.53	332.53
5,000	255.650	255.676	54,020	54,048	0.7364	0.7364	320.53	320.53
8,000	236.150	236.215	35,600	35,652	0.5252	0.5252	308.11	308.11
11,000	216.650	216.650	22,632	22,632	0.3639	0.3639	295.07	295.07
15,000	216.650	216.650	12,111	12,111	0.1948	0.1948	295.07	295.07
20,000	216.650	216.650	5,475	5,475	0.0880	0.0880	295.07	295.07
25,000	221.650	221.552	2,549	2,549	0.0401	0.0401	298.39	298.39
32,000	228.650	228.650	868.0	868.0	0.0132	0.0132	303.20	303.20

The model agrees with ICAO reference values to within rounding error in all layers, confirming correct implementation of the standard atmosphere equations.

17. Engine Modelling Methodology

17.1 Station Numbering Convention

All three engine architectures use the following consistent station numbering derived from ARP755A / SAE convention:

Table 17.1 Engine station numbering convention (all architectures)

Station	Location	Notes
0	Free-stream (far upstream)	Ambient ISA conditions
2	Compressor / Fan inlet	Post-intake; inlet recovery applied
3	Combustor inlet (compressor exit)	High-pressure compressor exit
4	Turbine inlet (combustor exit)	Turbine inlet temperature (TIT)
5	Turbine exit	High-pressure turbine exit for turbojet/turbofan; power turbine inlet for turboprop
7	Power turbine exit	Turboprop only
8	Nozzle exit	Core nozzle exit (all architectures)
18	Bypass nozzle exit	Turbofan bypass stream only

All intermediate thermodynamic states use total (stagnation) quantities T_t and P_t. Static conditions are computed only at the nozzle exit for thrust calculation. The subscript *t* denotes total, absent subscript denotes static.

> **Notation:** $\dot{m}$ — air mass flow rate [kg/s]; η_c — isentropic compressor efficiency; η_t — isentropic turbine efficiency; η_m — mechanical shaft efficiency; η_{gb} — gearbox efficiency (turboprop only); η_p — propeller efficiency (turboprop only); π_c — overall pressure ratio; π_f — fan pressure ratio; T_{t4} — turbine inlet temperature [K]; $h_f = 43.2$ MJ/kg — lower heating value of Jet-A fuel; $\gamma_{\text{air}} = 1.4$, $\gamma_{\text{gas}} = 1.33$, $c_{p,\text{air}} = 1004$ J/(kg·K), $c_{p,\text{gas}} = 1148$ J/(kg·K).

17.2 Turbojet Model

17.2.1 Inlet

Inlet total conditions at Station 2 (assuming inlet recovery $\eta_{\text{inlet}} \approx 1.0$ for subsonic, empirical ram recovery for supersonic):

$$T_{t2} = T_0\left(1 + \frac{\gamma - 1}{2} M^2\right) \quad [\text{K}]$$

$$P_{t2} = P_0\left(1 + \frac{\gamma - 1}{2} M^2\right)^{\gamma/(\gamma-1)} \cdot \eta_r(M) \quad [\text{Pa}]$$

where $\eta_r(M)$ is the ram pressure recovery factor (= 1.0 for $M \leq 1$; MIL-E-5007 formula for $M > 1$).

17.2.2 Compressor

$$T_{t3} = T_{t2}\left[1 + \frac{1}{\eta_c}\left(\pi_c^{(\gamma-1)/\gamma} - 1\right)\right] \quad [\text{K}]$$

$$P_{t3} = P_{t2} \cdot \pi_c \quad [\text{Pa}]$$

Compressor specific work: $W_c = c_{p,\text{air}}(T_{t3} - T_{t2})$ [J/kg]

17.2.3 Combustor

Burner pressure loss: $P_{t4} = P_{t3} \cdot (1 - \Delta P_{\text{burner}})$ where $\Delta P_{\text{burner}} \approx 0.05$ (5% pressure loss).

Fuel-air ratio from energy balance (including burner efficiency η_b):

$$f = \frac{c_{p,\text{gas}}\, T_{t4} - c_{p,\text{air}}\, T_{t3}}{\eta_b\, h_f - c_{p,\text{gas}}\, T_{t4}} \quad [-]$$

17.2.4 Turbine

Turbine work equals compressor work (shaft energy balance, η_m accounts for mechanical losses):

$$c_{p,\text{gas}}(T_{t4} - T_{t5}) = \frac{c_{p,\text{air}}\,(T_{t3} - T_{t2})}{\eta_m (1 + f)} \quad \Rightarrow \quad T_{t5} = T_{t4} - \frac{c_{p,\text{air}}(T_{t3} - T_{t2})}{\eta_m (1+f)\, c_{p,\text{gas}}}$$

$$P_{t5} = P_{t4}\left[1 - \frac{1}{\eta_t}\left(1 - \frac{T_{t5}}{T_{t4}}\right)\right]^{\gamma_g/(\gamma_g-1)} \quad [\text{Pa}]$$

17.2.5 Nozzle — Choked / Unchoked Logic

The critical nozzle pressure ratio (NPR) for a converging nozzle with $\gamma_g = 1.33$ is:

$$\text{NPR}_{\text{crit}} = \left(\frac{\gamma_g + 1}{2}\right)^{\gamma_g/(\gamma_g-1)} = \left(\frac{2.33}{2}\right)^{4.03} \approx 1.854$$

Compute actual NPR: $\text{NPR} = P_{t5}/P_0$.

Case 1 — Unchoked ($\text{NPR} \leq \text{NPR}_{\text{crit}}$):

$$P_e = P_0, \quad M_e = \sqrt{\frac{2}{\gamma_g - 1}\left[\left(\frac{P_{t5}}{P_0}\right)^{(\gamma_g-1)/\gamma_g} - 1\right]}$$

$$T_e = \frac{T_{t5}}{1 + \frac{\gamma_g - 1}{2} M_e^2}$$

Case 2 — Choked ($\mathrm{NPR} > \mathrm{NPR}_{\mathrm{crit}}$):

$$M_e = 1.0, \quad T_e = \frac{2T_{t5}}{\gamma_g + 1}, \quad P_e = \frac{P_{t5}}{\left(\frac{\gamma_g+1}{2}\right)^{\gamma_g/(\gamma_g-1)}}$$

17.2.6 Thrust and Efficiency

Exit velocity: $V_e = M_e\sqrt{\gamma_g\, R_{\text{gas}}\, T_e}$ [m/s], $R_{\text{gas}} = 287.4$ J/(kg·K)

Flight velocity: $V_0 = M \cdot a(H)$ [m/s]

Net specific thrust:

$$F_{\text{net}} = (1+f)V_e - V_0 + \frac{(P_e - P_0)A_e}{\dot{m}} \quad [\mathrm{N}/(\mathrm{kg/s})]$$

Net thrust (total): $F = \dot{m} \cdot F_{\text{net}}$ [N]

TSFC: $\mathrm{TSFC} = \frac{\dot{m}_f}{F} = \frac{f}{F_{\text{net}}}$ [kg/(N·s)]

Thermal efficiency: $\eta_{\text{th}} = \frac{(1+f)V_e^2 - V_0^2}{2fh_f}$

Propulsive efficiency: $\eta_{\text{pr}} = \frac{2F_{\text{net}}V_0}{(1+f)V_e^2 - V_0^2 + \text{pressure terms}}$

Overall efficiency: $\eta_o = \eta_{\text{th}} \times \eta_{\text{pr}}$

17.2.7 Turbojet Model Assumptions

Table 17.2 Turbojet model assumptions and justification

Assumption	Value / Treatment	Justification
Inlet recovery η_r	1.0 for M ≤ 1; MIL-E-5007 for M > 1	Standard intake assumption; no inlet geometry modelled
Burner pressure loss	5%	Typical for annular combustors; ±1% sensitivity documented
Burner efficiency η_b	0.99	Conservative estimate; modern annular combustors > 0.98
Mechanical efficiency η_m	0.99	Bearing and seal losses
Calorically perfect gas	γ, c_p constant per stream	Valid for steady-state, fixed throttle; major simplification accepted
Nozzle type	Converging-only	Converging-diverging nozzle (J79, Atar 9C with reheat) treated as converging-only in Phase 1

17.3 Turbofan Model — Separate-Exhaust Architecture

> **Architecture Choice:** The Phase 1 turbofan model uses a **separate-exhaust (unmixed) configuration**: the bypass stream and core stream have independent nozzles and do not mix before exhausting. This configuration is simpler than a mixed-exhaust model, analytically transparent, and covers both high-BPR commercial and low-BPR military turbofans with the same framework. The same generalized equations apply to CFM56-5B4, CF6-80C2, F100-PW-229, M88-2, and Kaveri (estimated) with different architecture parameters.

17.3.1 Mass Flow Split

Bypass ratio: $\mathrm{BPR} = \alpha = \dot{m}_{\text{bypass}} / \dot{m}_{\text{core}}$

Core mass flow: $\dot{m}_c = \dot{m}_{\text{total}} / (1 + \alpha)$

Bypass mass flow: $\dot{m}_b = \alpha \cdot \dot{m}_c$

17.3.2 Fan

Fan inlet conditions: same as Station 2 (both streams see the fan).

$$T_{t2.5,b} = T_{t2}\left[1 + \frac{1}{\eta_f}\left(\pi_f^{(\gamma-1)/\gamma} - 1\right)\right] \quad [\mathrm{K}]$$

$$P_{t2.5} = P_{t2} \cdot \pi_f \quad [\mathrm{Pa}]$$

Fan specific work: $W_f = c_{p,\text{air}}(T_{t2.5,b} - T_{t2})$ [J/kg]

17.3.3 Core Compressor

Core compressor pressure ratio: $\pi_{c,\text{core}}$ (inlet at fan exit, $P_{t3} = P_{t2.5} \cdot \pi_{c,\text{core}}$)

$$T_{t3} = T_{t2.5,b}\left[1 + \frac{1}{\eta_c}\left(\pi_{c,\text{core}}^{(\gamma-1)/\gamma} - 1\right)\right]$$

Overall OPR: $\pi_{\text{OPR}} = \pi_f \times \pi_{c,\text{core}}$

17.3.4 Combustor and Turbine (Core)

Identical to turbojet §17.2.3–17.2.4 applied to core stream.

Turbine work balance including fan drive (single spool assumed for simplicity in Phase 1 model; two-spool effects captured through separate compressor efficiency inputs):

$$c_{p,\text{gas}}(T_{t4} - T_{t5}) = \frac{(1+\alpha)\,c_{p,\text{air}}(W_f) + c_{p,\text{air}}(T_{t3} - T_{t2.5,b})}{\eta_m (1 + f)}$$

17.3.5 Core Nozzle (Station 5 → 8)

Apply choked/unchoked logic from §17.2.5 with $\mathrm{NPR} = P_{t5}/P_0$.

Core nozzle thrust: $F_{\text{core}} = \dot{m}_c\left[(1+f)V_{e8} - V_0 + \frac{(P_{e8}-P_0)A_8}{\dot{m}_c}\right]$

17.3.6 Bypass Nozzle (Station 2.5 → 18)

Bypass stream is cold air ($\gamma = 1.4$).

Apply choked/unchoked logic: $\mathrm{NPR}_b = P_{t2.5}/P_0$. For typical turbofans $\mathrm{NPR}_b < 1.854$, so bypass nozzle is unchoked; nonetheless the check is always performed.

$$V_{e18} = \sqrt{2c_{p,\text{air}}T_{t2.5,b}\left[1 - \left(\frac{P_0}{P_{t2.5}}\right)^{(\gamma-1)/\gamma}\right]} \quad [\mathrm{m/s}]$$

Bypass nozzle thrust: $F_{\text{bypass}} = \dot{m}_b(V_{e18} - V_0)$

17.3.7 Total Thrust and TSFC

$$F_{\text{total}} = F_{\text{core}} + F_{\text{bypass}} \quad [\mathrm{N}]$$

$$\mathrm{TSFC} = \frac{\dot{m}_f}{F_{\text{total}}} = \frac{\dot{m}_c \cdot f}{F_{\text{total}}}$$

Units: kg/(N·s)

> **Note on Low-BPR vs High-BPR Fidelity:** The same separate-exhaust model applies to all five turbofan engines. However, for low-BPR military turbofans (F100-PW-229, M88-2, Kaveri), the fan-core interaction and turbine cooling effects are more significant. Model accuracy is expected to be lower for military turbofans than for high-BPR commercial engines. All fidelity differences are documented in §23.

17.4 Turboprop Model

17.4.1 Gas Generator

The turboprop gas generator follows the same thermodynamic path as the turbojet up to Station 5 (turbine exit). At Station 5, instead of all energy going to a nozzle, most shaft power is extracted by a **power turbine** and transmitted through a gearbox to the propeller.

17.4.2 Power Turbine

The power turbine expands gas from P_{t5} to a residual back-pressure slightly above P_0:

$$T_{t7} = T_{t5} - \eta_{t,\text{pt}}\,T_{t5}\left[1 - \left(\frac{P_{t7}}{P_{t5}}\right)^{(\gamma_g-1)/\gamma_g}\right]$$

Power turbine shaft power per unit core mass flow:

$$\dot{W}_{\text{pt}} = c_{p,\text{gas}}(T_{t5} - T_{t7}) \quad [\text{J/kg}]$$

17.4.3 Shaft Power, Gearbox, Propeller

Shaft power at gearbox input (mechanical efficiency η_m):

$$P_{\text{shaft}} = \dot{m}_c \cdot \dot{W}_{\text{pt}} \cdot \eta_m \quad [\text{W}]$$

Power at propeller shaft after gearbox (gearbox efficiency η_{gb}):

$$P_{\text{prop}} = P_{\text{shaft}} \cdot \eta_{\text{gb}} \quad [\text{W}]$$

Propeller thrust from shaft power (propeller efficiency η_p):

$$F_{\text{prop}} = \frac{\eta_p \cdot P_{\text{prop}}}{V_0} \quad [\text{N}] \quad (V_0 > 0)$$

> **Static Thrust Case ($V_0 = 0$, i.e., $M = 0$):** The expression $\eta_p P_{\text{prop}} / V_0$ is indeterminate at $V_0 = 0$. At static conditions, propeller thrust is computed from actuator disc theory:
>
> $$F_{\text{prop,static}} = \left(2\,\rho_0\,A_{\text{disc}}\,(\eta_p P_{\text{prop}})^2\right)^{1/3}$$
>
> where $A_{\text{disc}} = \pi D_p^2/4$ is the propeller disc area and D_p is the propeller diameter. This requires knowledge of propeller diameter (available for all three turboprop engines in §9). The model smoothly transitions between the static and forward-flight expressions using a velocity blending function for $M \in [0,\, 0.05]$.

17.4.4 Residual Jet Thrust

Gas exhausted from the power turbine at Station 7 passes through a short exhaust duct and nozzle. Residual jet velocity:

$$V_{e8} = \sqrt{2\,c_{p,\text{gas}}\,T_{t7}\left[1 - \left(\frac{P_0}{P_{t7}}\right)^{(\gamma_g - 1)/\gamma_g}\right]}$$

Residual jet thrust: $F_{\text{jet}} = \dot{m}_c (V_{e8} - V_0)$

17.4.5 Equivalent Shaft Horsepower (ESHP) and Total Thrust

Total effective thrust:

$$F_{\text{total}} = F_{\text{prop}} + F_{\text{jet}} \quad [\text{N}]$$

Equivalent shaft horsepower (ESHP) — useful for comparing shaft-based and thrust-based performance:

$$\text{ESHP} = \frac{P_{\text{prop}} + F_{\text{jet}} \cdot V_0}{\eta_p \cdot 745.7} \quad [\text{hp}]$$

Brake Specific Fuel Consumption:

$$\text{BSFC} = \frac{\dot{m}_f}{P_{\text{shaft}}} \quad \text{(SI)} = \frac{\dot{m}_f \times 2.685 \times 10^6}{P_{\text{shaft}}} \quad \text{(Imperial)}$$

SI units: kg/(W·s); Imperial: lb/(hp·h)

17.4.6 Turboprop Model Assumptions

Table 17.3 Turboprop model assumptions

Parameter	Value / Treatment
Gearbox efficiency η_{gb}	0.98 (typical for modern reduction gearboxes; PT6A, PW127G range 0.97–0.99)
Propeller efficiency η_p	Function of advance ratio $J = V_0/(nD_p)$; simplified as 0.80 at cruise, 0.70 at climb, for Phase 1; propeller map not modelled
Power turbine exit pressure P_{t7}	Assumed equal to P_0 (complete expansion in power turbine); corrected for actual back-pressure where needed
Static thrust	Actuator disc approximation (see §17.4.3)
Valid Mach range	0.0–0.7 (consistent with turboprop operational envelope)

17.5 Model Comparison Summary

Table 17.4 Summary of engine model scope and fidelity

Model	Architecture	Engines	Valid M Range	Known Limitations
Turbojet	Single-spool	J79, Atar 9C, R-25	0.0–2.5	No convergent-divergent nozzle; ram recovery model approximate
Turbofan (Sep.)	Two-stream	CFM56-5B4, CF6-80C2, F100-PW-229, M88-2, Kaveri	0.0–0.92 HBP; 0.0–1.8 LBP	No mixing; turbine cooling not modelled; Kaveri data estimated
Turboprop	Free power turbine	PT6A-67D, PW127G, AE 2100D3	0.0–0.7	Propeller map simplified; blade pitch scheduling not modelled

18. Model Validity Checks

Every engine model (turbojet, turbofan, turboprop) must perform a comprehensive set of validity checks before executing the thermodynamic cycle. The Digital Twin Coordinator (§19) enforces these checks and returns structured error/warning codes before model evaluation proceeds. This section defines the complete set of required checks per functional requirement FR-07.

18.1 Input Range Checks

Table 18.1 Input range validity checks — all engine architectures

Parameter	Valid Range	Error Code	Action
Geopotential altitude H	0 m ≤ H ≤ 32,000 m	ERR-ATM-01	Reject; return NaN
Mach number M — Turbojet	0.0 ≤ M ≤ 2.5	ERR-TJ-01	Reject; return NaN
Mach number M — Turbofan HBP	0.0 ≤ M ≤ 0.92	ERR-TF-01	Reject; return NaN
Mach number M — Turbofan LBP	0.0 ≤ M ≤ 1.8	ERR-TF-02	Reject; return NaN
Mach number M — Turboprop	0.0 ≤ M ≤ 0.7	ERR-TP-01	Reject; return NaN
Throttle fraction θ	0.0 ≤ θ ≤ 1.0	ERR-IN-01	Reject; return NaN
Turbine inlet temperature T_{t4}	800 K ≤ T_{t4} ≤ 2200 K	ERR-IN-02	Warn if outside; reject if T_{t4} < 400 K or > 2500 K
Overall pressure ratio π_c	1.0 < π_c ≤ 50	ERR-IN-03	Reject; physically impossible

18.2 Component Efficiency Checks

Table 18.2 Component efficiency validity checks

Parameter	Valid Range	Error Code	Rationale
Compressor isentropic efficiency η_c	0.70 ≤ η_c ≤ 0.92	ERR-COMP-01	Physical range for axial/centrifugal compressors
Turbine isentropic efficiency η_t	0.85 ≤ η_t ≤ 0.95	ERR-TURB-01	Physical range for cooled/uncooled turbines
Fan isentropic efficiency η_f	0.80 ≤ η_f ≤ 0.92	ERR-FAN-01	Turbofan only
Burner efficiency η_b	0.95 ≤ η_b ≤ 1.0	ERR-BURN-01	Modern combustors
Mechanical efficiency η_m	0.97 ≤ η_m ≤ 1.0	ERR-MECH-01	Shaft bearing losses

Gearbox efficiency η_{gb}	0.96 ≤ η_gb ≤ 1.0	ERR-GB-01	Turboprop only
Propeller efficiency η_p	0.0 < η_p ≤ 0.92	ERR-PROP-01	η_p = 0.0 only at static condition (handled separately)

18.3 Thermodynamic Consistency Checks

After each thermodynamic step, the following checks are applied to detect physically impossible outputs before proceeding to the next component:

Table 18.3 Thermodynamic consistency checks (applied sequentially per step)

Check	Condition Verified	Error Code	Action
Non-negative mass flow	$\dot{m} > 0$ at every station	ERR-FLOW-01	Fatal; abort cycle
Temperature increase across combustor	$T_{t4} > T_{t3}$	ERR-COMB-01	Fatal; abort cycle
Temperature decrease across turbine	$T_{t5} < T_{t4}$	ERR-TURB-02	Fatal; abort cycle
Pressure decrease across turbine	$P_{t5} < P_{t4}$	ERR-TURB-03	Fatal; abort cycle
Positive fuel-air ratio	$f > 0$	ERR-FAR-01	Fatal; abort cycle
Physically bounded fuel-air ratio	$f < 0.10$ (stoichiometric limit for Jet-A)	ERR-FAR-02	Warn; TIT likely too high for stated η_b
Positive exit velocity	$V_e > 0$	ERR-NOZ-01	Fatal; abort cycle
Real-valued exit Mach	$P_{t5}/P_0 \geq 1.0$ (nozzle can expand)	ERR-NOZ-02	Fatal; abort cycle
No NaN / Inf in output vector	All outputs finite and real	ERR-NAN-01	Fatal; abort; log all inputs

18.4 Nozzle Condition Check

The nozzle check is performed explicitly before exit conditions are computed:

1. Compute NPR = P_t5 / P_0.
2. Compare NPR against NPR_crit = 1.854 (γ_g = 1.33) or 1.893 (γ_g = 1.4, bypass stream).
3. Set nozzle state flag: `CHOKED` or `UNCHOKED`.

4. Log nozzle state to Simulation_Log for every operating point.
5. If NPR < 1.0: fatal error ERR-NOZ-02 (pressure ratio below unity — impossible).

18.5 Out-of-Range Warning System

The validity check system distinguishes between hard errors (computation aborted, NaN returned) and soft warnings (computation proceeds with a warning flag visible in GUI and logged to database):

Table 18.4 Warning vs error classification

Severity	Condition	GUI Display	DB Action
Fatal Error	Any ERR-*-0x condition above	Red banner: "Invalid operating condition — computation aborted"	Log with error code; result = NaN
Warning	f > 0.07, T_{t4} > 1900 K (estimated), efficiency at boundary	Amber banner: "Operating near model limits — results may be less accurate"	Log warning flag; result stored
Kaveri Disclaimer	Any Kaveri GTX-35VS computation	Yellow box: "ESTIMATED / ACADEMIC MODEL — NOT MANUFACTURER-CONFIRMED PERFORMANCE"	Data_Type = 'Simulated_Estimated' in Performance_Data
ML Fallback	ML surrogate unavailable	Info banner: "ML surrogate not available — physics model used"	Prediction_Source = 'Physics'

18.6 Model Validity Summary by Engine

Table 18.5 Validity check status per engine

Engine	Arch.	M Range	H Range (m)	Special Checks
J79-GE-17	TJ	0.0–2.0	0–18,000	Afterburner flag; reheat TIT limit
Atar 9C	TJ	0.0–2.2	0–18,000	Afterburner flag
R-25-300	TJ	0.0–2.5	0–20,000	Two-spool flag (treated as single for Phase 1)
CFM56-5B4	TF-HBP	0.0–0.92	0–13,000	High-altitude idle check; ice crystal ingestion warning (not modelled)
CF6-80C2B1F	TF-HBP	0.0–0.90	0–13,000	Same as CFM56
F100-PW-229	TF-LBP	0.0–1.8	0–15,000	Afterburner flag; LBP nozzle choking

M88-2	TF-LBP	0.0–1.8	0–17,000	Afterburner flag
Kaveri GTX-35VS	TF-LBP	0.0–1.8*	0–15,000*	Estimated data; Kaveri disclaimer mandatory; all outputs flagged ESTIMATED
PT6A-67D	TP	0.0–0.55	0–9,000	Static thrust actuator disc; propeller torque limit
PW127G	TP	0.0–0.65	0–9,000	Static thrust actuator disc; PW127G rated ceiling 25,000 ft ≈ 7,620 m ISA
AE 2100D3	TP	0.0–0.70	0–9,000	6-blade composite propeller; blade pitch scheduling not modelled

* Kaveri operational envelope at low confidence; values marked with asterisk are based on estimated academic parameters, not manufacturer-confirmed specifications.

19. Digital Twin Architecture

> **Academic Scope Disclaimer:** The Digital Twin described in this document is an *academic steady-state software Digital Twin*. It is a structured set of MATLAB computational models that mirror an engine's thermodynamic performance at discrete operating points. It is NOT: a real-time physical engine Digital Twin; a certified engineering or flight-safety Digital Twin; a flight-control or hardware-in-the-loop system; nor a replacement for type-certified engine simulation or OEM performance tools. Results are intended for comparative academic analysis and engineering trend studies only.

19.1 Five-Level Digital Twin Architecture

The Digital Twin is structured as five progressive fidelity levels (L1–L5). Each level adds capability while fully encompassing the capabilities of all lower levels. The Digital Twin Coordinator MATLAB class orchestrates all levels and enforces the validity checks defined in §18.

Table 19.1 Five-level Digital Twin architecture

Level	Name	Description	Inputs	Outputs	MATLAB Implementation
L1	Static Physics Model	Steady-state thermodynamic cycle at one design-point condition. Single operating point per evaluation.	Engine ID, M, H, θ (throttle)	F [N], TSFC [kg/(N·s)], T_{t4} [K], η_th, η_pr, η_o, station temperatures and pressures	`TJ_Model.m`, `TF_Model.m`, `TP_Model.m`

L2	Altitude Parametric Model	L1 repeated across the full altitude range (0–32,000 m) at fixed M and θ, producing performance-vs-altitude curves.	Engine ID, M range, H vector, θ	F(H), TSFC(H), η(H) curves; plotted performance envelopes	`AltitudeSweep.m`
L3	Multi-Condition Performance Map	L1 evaluated over the full (M, H, θ) grid to generate complete engine performance maps for cruise, climb, takeoff, and combat conditions.	Engine ID, M×H×θ grid	3D performance map arrays; contour plots; thrust/TSFC maps over (M, H) at fixed throttle	`PerformanceMap.m`
L4	Database-Connected Model	L3 results are stored to SQLite via the MATLAB Database Toolbox. Reference data retrieved for overlay. Parameter provenance tracked via Parameter_Provenance table.	Engine ID, operating condition grid	Populated Simulation_Log and Performance_Data tables; reference overlay plots	`DTwin_L4.m`, `DB_Interface.m`
L5	ML-Augmented Model	ML surrogate models (§20) are trained on L3 physics-generated data. L5 routes rapid-query requests through the ML surrogate with physics fallback. Surrogate fidelity relative to L3 physics model is reported.	Engine ID, M, H, θ (rapid-query path)	ML predictions with R^2, RMSE; physics vs ML comparison plots; prediction latency report	`ML_Surrogate.m`, `DTwin_L5.m`

19.2 Digital Twin Coordinator

The Digital Twin Coordinator (`DT_Coordinator.m`) is the central orchestration class. Its responsibilities are:

1. **Input validation:** Applies all checks from §18 before dispatching to any physics model.
2. **Model routing:** Selects the appropriate engine model (TJ/TF/TP) based on Engine_ID lookup from the database.
3. **Level selection:** Executes L1 through L5 computations on demand.
4. **Surrogate routing (L5):** If trained ML surrogate is available and query is within training domain, returns ML prediction. Otherwise falls back to L4 physics with GUI warning.
5. **Result logging:** Writes all results (physics and ML) to Performance_Data and Simulation_Log with correct Data_Type and Prediction_Source fields.

6. **Warning propagation:** Forwards all error codes and Kaveri disclaimers to the GUI layer.

19.3 Data Flow Architecture

The data flow between system components is as follows:

Table 19.2 Data flow between Digital Twin components

Source	Data	Destination	Interface
GUI (App Designer)	Engine selection, M, H, θ, mission profile	DT_Coordinator	Function call / callback
DT_Coordinator	Validated inputs	TJ/TF/TP physics model	Function call
Physics model	Thermodynamic output struct	DT_Coordinator → GUI	Return struct
DT_Coordinator	Simulation results	SQLite DB (Performance_Data, Simulation_Log)	MATLAB Database Toolbox SQL INSERT
SQLite DB	Reference data, engine parameters, provenance	DT_Coordinator	SQL SELECT (parameterized)
ML_Surrogate	Predicted performance vector	DT_Coordinator → GUI	Surrogate function call

19.4 Fidelity and Limitations

The five-level DT achieves progressive fidelity through systematic parametric sweeps. The following limitations apply to all levels:

- All computations are **steady-state**. Transient engine behaviour (acceleration, deceleration, surge recovery, bleed scheduling) is not modelled at any level.
- Component maps (compressor/turbine characteristic maps) are not used. Component performance is represented by single-point isentropic efficiency values.
- Turbine cooling is not modelled. This introduces systematic error at high TIT conditions.
- Installation effects (nacelle drag, bleed extraction for aircraft systems) are not modelled.
- The DT does not interface with real hardware at any level. It is purely software-based.
- L5 ML augmentation improves query speed but does not add physical insight beyond L3.

20. Machine Learning Architecture

> **Clarification on ML Validation Scope:** All ML surrogates in this project are trained on physics-model-generated data (L3 outputs). Therefore, ML validation metrics (R^2, RMSE, MAE, MAPE)

> measure how well the ML model reproduces the *physics model* — not how well it predicts *real engine performance*. The distinction is explicitly maintained throughout. "ML vs reference engine data" comparisons are not performed; only "ML vs held-out physics data" comparisons are used for ML acceptance criteria.

20.1 Training Data Generation

Training data is generated by running the validated physics models (DT L3) over the engine-type-specific operating grids defined below. Invalid conditions are excluded by the model validity checks (§18) *before* data is stored — no invalid-region data is added merely to increase dataset size.

Table 20.1 ML training data grid specification by engine architecture

Architecture	Altitude H [m]	Mach M [-]	Throttle θ [-]	Points/Engine	Total (3+5+3 engines)
Turbojet (TJ)	21 levels: 0, 1,000, 2,000, … 20,000 m	9 levels: 0.0, 0.3, 0.6, 0.9, 1.2, 1.5, 1.8, 2.1, 2.5	6 levels: 0.5, 0.6, 0.7, 0.8, 0.9, 1.0	Up to 1,134	3 × 1,134 (after filter)
Turbofan HBP (CFM56, CF6)	21 levels: 0–13,000 m + cruise altitudes	8 levels: 0.0, 0.15, 0.3, 0.5, 0.65, 0.78, 0.85, 0.92	6 levels: 0.5, 0.6, 0.7, 0.8, 0.9, 1.0	Up to 1,008	2 × 1,008 (after filter)
Turbofan LBP (F100, M88, Kaveri)	21 levels: 0–17,000 m	9 levels: 0.0, 0.3, 0.6, 0.9, 1.2, 1.5, 1.8, 2.0, 2.5*	6 levels: 0.5, 0.6, 0.7, 0.8, 0.9, 1.0	Up to 1,134	3 × 1,134 (after filter)
Turboprop (PT6A, PW127G, AE 2100D3)	21 levels: 0–9,000 m	8 levels: 0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7	6 levels: 0.5, 0.6, 0.7, 0.8, 0.9, 1.0	Up to 1,008	3 × 1,008 (after filter)

* Turbojet/LBP turbofan Mach range extended to physics model validity limit. Kaveri training data generated with estimated engine parameters and flagged ESTIMATED throughout.

20.2 Architecture-Specific Surrogate Strategy

Separate surrogate models are trained per engine *architecture class* (not per individual engine). A single cross-architecture model is not used because turboprop, turbofan, and turbojet performance curves have fundamentally different functional forms and Mach ranges. Engine type and architecture are encoded as categorical features if within-architecture multi-engine surrogates are trained.

Table 20.2 Surrogate model strategy by architecture

Architecture Class	Surrogate Scope	Feature Vector	Target Variables
Turbojet	One surrogate per engine (J79, Atar 9C, R-25)	[M, H, θ]	[F_net, TSFC, η_th, η_pr]
Turbofan HBP	One surrogate per engine (CFM56-5B4, CF6-80C2)	[M, H, θ]	[F_total, TSFC, η_th]
Turbofan LBP	One surrogate per engine (F100, M88, Kaveri*)	[M, H, θ]	[F_total, TSFC, η_th]
Turboprop	One surrogate per engine (PT6A-67D, PW127G, AE 2100D3)	[M, H, θ]	[F_total, BSFC, P_shaft, ESHP]

20.3 ML Model Selection

Four surrogate model families are evaluated for each engine. The best-performing model (highest R^2 on held-out physics data) is selected as the operational surrogate for L5:

Table 20.3 ML model candidates evaluated per engine

Model	MATLAB Function	Key Hyperparameters	Expected Strengths
Polynomial Regression (Degree 3–5)	`fitrlinear` / `polyfitn`	Degree, regularisation λ	Fast; interpretable; good for smooth physics curves
Random Forest	`TreeBagger` (Statistics and ML Toolbox)	NumTrees (100–500), MinLeafSize	Non-parametric; handles local nonlinearities
Gaussian Process Regression (GPR)	`fitrgp`	Kernel function (RBF / Matérn 5/2), σ_n	Provides uncertainty estimates; excellent for smooth interpolation
Artificial Neural Network (ANN)	`trainnet` / `fitnet`	Layers: [64, 32, 16]; activation: ReLU; dropout 0.1	High capacity; best for complex multi-output mappings

20.4 Dataset Split

Training data (physics-model outputs, after invalid-condition filtering) is split as follows:

Training: 70% | Validation: 15% | Test (held-out): 15%

The split is **stratified by (altitude bin, Mach bin)** to ensure all regions of the operating envelope are represented in all three splits. Random splitting without stratification is explicitly prohibited — it could result in the test set containing only low-altitude, low-Mach conditions that are easier to predict, producing artificially optimistic metrics.

20.5 Acceptance Criteria

> **Important Clarification:** $R^2 \geq 0.99$ is a *target* for well-behaved physics-model surrogates, not a guaranteed outcome. It indicates that the ML model reproduces the physics model to within 1% variance. It does NOT mean the physics model itself matches real engine data to 99% accuracy. If a model does not meet the threshold, the actual result is reported, root causes are investigated, models are compared, and the limitation is documented. The dataset or model is not manipulated to force $R^2 \geq 0.99$.

Table 20.4 ML surrogate acceptance criteria

Metric	Target	Minimum Acceptable	Action if Below Minimum
R^2 (coefficient of determination)	≥ 0.99	≥ 0.95	Report actual; investigate feature engineering; document limitation
RMSE (root mean square error)	< 0.5% of mean target value	< 2% of mean target value	Report actual RMSE; identify high-error regions
MAE (mean absolute error)	< 0.3% of mean	< 1.5% of mean	As above
MAPE (mean absolute percentage error)	< 1%	< 5%	Report; investigate region of high percentage error
Prediction time per point	< 1 ms	< 10 ms	Justified if GPR/ANN; note in §27 deliverable

20.6 Physics Validation vs ML Validation — Explicit Separation

The following two validation steps are entirely separate and must not be conflated:

Table 20.5 Physics model validation vs ML surrogate validation — explicit separation

Validation Type	Comparison	Data Source	Reported Metric	Section
A. Physics Model Validation	Physics model output vs published/reference engine data	Jane's, EASA TCDS, AFSC reports, academic literature (per §15)	Thrust error %, TSFC error %, or BSFC error % at specified reference conditions	§23
B. ML Surrogate Validation	ML prediction vs held-out physics-model data (15% test split)	Physics model L3 outputs (not reference engine data)	R^2, RMSE, MAE, MAPE (Table 20.4)	§23 + §27

ML surrogate performance (Type B) does not constitute validation of the physics model against real engines (Type A). Both validations must be completed and reported separately.

21. Engine Selection Optimisation

21.1 Two-Stage Selection Algorithm

The engine selection algorithm operates in two strictly sequential stages. Hard constraints are applied first; only engines that pass all constraints proceed to weighted scoring. This design ensures that a high weighted score cannot override a constraint violation.

Stage 1 — Hard Constraint Filtering

Each mission profile (§21.4) defines a set of hard constraints. An engine is admitted to Stage 2 only if it satisfies *all* of the following:

Table 21.1 Hard constraint categories

Constraint	Parameter	Test	Failure Action
Minimum thrust	F_min [kN]	F(M_cruise, H_cruise, θ=1.0) ≥ F_min	Eliminated from ranking
Maximum TSFC / BSFC	TSFC_max [kg/(N·s)]	TSFC at cruise ≤ TSFC_max	Eliminated
Altitude ceiling	H_ceil [m]	Engine must produce positive net thrust at H_ceil	Eliminated
Architecture compatibility	Engine type flag	Mission architecture (TJ / TF / TP) must match engine type	Eliminated
Maximum engine dry weight	W_max [kg]	Engine dry weight ≤ W_max (mission installation limit)	Eliminated

After Stage 1 filtering, three outcomes are possible:

- **Empty feasible set:** No engine passes all constraints. The GUI displays: "No engine in the database meets all hard constraints for this mission profile. Relax one or more constraints to proceed." No ranking is produced.
- **Single engine in feasible set:** One engine passes all constraints. It is reported as the only feasible selection with a note: "Only one engine meets all hard constraints — no meaningful comparative ranking is possible." A full performance report is still generated for that engine.
- **Two or more engines in feasible set:** Normal Stage 2 weighted scoring proceeds.

Stage 2 — Weighted Scoring

For each engine i in the feasible set, a composite score S_i is computed using min-max normalised performance metrics:

$$S_i = w_1 \tilde{F}_i + w_2 \tilde{\eta}_i + w_3 \widetilde{\text{TSFC}}_i^{-} + w_4 \tilde{H}_i + w_5 \widetilde{\text{TWR}}_i$$

where $\sum_{k=1}^{5} w_k = 1$ and all weights $w_k \geq 0$.

21.2 Normalisation

All five performance metrics are min-max normalised to the range [0, 1] before scoring. Higher score is always better:

Higher-is-better metrics (thrust F, efficiency η, altitude ceiling H, TWR):

$$\tilde{x}_i = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}}$$

Lower-is-better metrics (TSFC, BSFC, fuel consumption):

$$\tilde{x}_i^{-} = \frac{x_{\max} - x_i}{x_{\max} - x_{\min}}$$

Degenerate case ($x_{\max} = x_{\min}$, all feasible engines have identical value):

$$\tilde{x}_i = 0.5 \quad \text{for all } i \quad \text{(tie; this metric provides no discrimination)}$$

This case is logged to the GUI with the message: "All feasible engines have identical [metric] — this criterion does not differentiate candidates."

21.3 Thrust-to-Weight Ratio

The thrust-to-weight ratio (TWR) used in scoring is defined as the engine-level TWR based on engine dry weight only:

$$\text{TWR} = \frac{F_{\text{rated}}}{W_{\text{engine,dry}} \times g_0}$$

where F_{rated} is the sea-level static maximum dry thrust [N], $W_{\text{engine,dry}}$ is the dry engine mass [kg] (from the Engines table), and $g_0 = 9.80665$ m/s². Aircraft-level TWR (which depends on airframe mass, fuel load, and lift) is not computed in Phase 1 and this limitation is documented in §26.

21.4 Mission Presets

Six standard mission profiles are pre-loaded in the database (Mission_Profiles table). Users may also define custom profiles in the GUI:

Table 21.2 Standard mission profile presets

Mission ID	Name	Architecture	Cruise M	Cruise H [m]	F_min [kN]	Priority Weights (F, η, TSFC, H, TWR)
M001	Subsonic Commercial Transport	TF-HBP	0.82	11,000	60	0.15, 0.30, 0.35, 0.10, 0.10
M002	Air Superiority Fighter	TJ / TF-LBP	1.2	12,000	70	0.40, 0.05, 0.10, 0.20, 0.25
M003	Ground Attack / CAS	TJ / TF-LBP	0.85	5,000	50	0.35, 0.05, 0.15, 0.15, 0.30
M004	Maritime Patrol / ISR	TP	0.45	3,000	15	0.10, 0.20, 0.40, 0.10, 0.20
M005	STOL Transport	TP	0.35	2,000	20	0.25, 0.15, 0.25, 0.10, 0.25
M006	High-Altitude Reconnaissance	TJ / TF-LBP	0.80	20,000	20	0.15, 0.20, 0.25, 0.35, 0.05

21.5 MATLAB Implementation Notes

The selection algorithm is implemented in `EngineSelector.m`. Key implementation requirements:

- Hard constraint filtering is a separate function `apply_hard_constraints(engines, mission)` called before scoring.
- Normalisation is computed from the feasible set only (not all 11 engines), ensuring ties and outliers in eliminated engines do not affect normalisation bounds.
- Division-by-zero in normalisation is explicitly handled with the tie-value case.
- Selection results are written to Simulation_Log with mission profile ID, ranking, and per-metric scores.
- GUI displays the full ranking table with colour-coded scores and individual metric breakdown.

22. GUI Architecture

22.1 Implementation Platform

The GUI is implemented in **MATLAB R2023b App Designer** as a single `.mlapp` file (`EngineAnalyser_v1.mlapp`). App Designer is the standard MATLAB GUI framework for R2019b+. The application requires: MATLAB R2023b; Statistics and Machine Learning Toolbox; Database Toolbox; Optimization Toolbox (optional, for constraint solving). No additional third-party toolboxes are required.

22.2 Eight-Tab Layout

Table 22.1 GUI tab specification

Tab #	Tab Name	Function	Key Controls
1	Engine Database Browser	View all 11 engines; filter by architecture; display parameter provenance and confidence	Architecture dropdown; engine listbox; parameter table with colour-coded confidence; source hyperlinks
2	Single-Point Analysis	Run L1 physics model at one (M, H, θ) point for a selected engine	Engine dropdown; M, H, θ sliders with units and valid range annotations; "Run" button; output panel with station temperatures/pressures; warning banner area
3	Performance Envelope (L2/L3)	Generate F(H), TSFC(H), η(H) curves and full 3D performance maps	Engine selection; M range; H range; θ setting; axis selectors; export to CSV button
4	Digital Twin Comparison	Overlay reference data vs simulated vs ML-predicted performance for any engine	Engine selection; overlay selector (Reference / Simulated / ML); confidence error bar display; source labels; Kaveri disclaimer panel if applicable
5	ML Surrogate Manager	Train, evaluate, and query ML surrogates	Engine selection; model type selector (Poly/RF/GPR/ANN); train/evaluate buttons; R^2, RMSE, MAE, MAPE display; prediction time display; surrogate availability indicator
6	Mission-Based Engine Selection	Run two-stage selection algorithm for a mission profile	Mission preset dropdown; custom constraint inputs; weight sliders (sum enforced = 1.0); "Select" button; ranked results table; elimination reason display
7	Validation Report	Display physics model validation results against reference data	Engine selector; validation point table (from §23); model vs reference error display; error bar plots; source confidence labels
8	Settings and Export	Database connection; unit preferences; export options	DB path field; unit system toggle (SI/Imperial); plot export (PNG/PDF); data export (CSV/MAT); model version display

22.3 Mandatory GUI Warnings and Indicators

The following warnings and status indicators are mandatory in all tabs where relevant. They cannot be disabled by the user:

Table 22.2 Mandatory GUI warning and indicator requirements

Condition	Display Location	Message	Colour
Operating outside model's valid Mach range	Tab 2, Tab 3, top banner	"Warning: M = [x] exceeds valid range [M_min, M_max] for [engine] — results not available"	Red
Operating outside model's valid altitude range	Tab 2, Tab 3, top banner	"Warning: H = [x] m exceeds valid range [0, H_max] m for [engine]"	Red
Estimated engine data in use (Kaveri)	Tab 2, Tab 4, Tab 6, dedicated panel	"ESTIMATED / ACADEMIC MODEL — NOT MANUFACTURER-CONFIRMED PERFORMANCE"	Amber (permanent)
ML surrogate not available	Tab 4, Tab 5, status label	"ML surrogate not trained — physics model result displayed"	Blue/info
Model is steady-state only	Tab 2 footer (permanent)	"Steady-state model only — transient dynamics not represented"	Grey (permanent)
Low-confidence reference data	Tab 4, Tab 7, data point labels	[Source name] tagged with confidence: HIGH / MED / LOW (colour-coded)	Green/Amber/Red
Hard constraint violation in mission selection	Tab 6, results table	Engine row greyed out with constraint name displayed in "Eliminated — [constraint]" column	Grey (eliminated)
Empty feasible set in mission selection	Tab 6, results area	"No engine meets all hard constraints for this mission. Relax constraints to proceed."	Red banner

22.4 Units and Display

All GUI inputs and outputs display units next to the value. The Settings tab (Tab 8) allows switching between SI and mixed-Imperial units. Conversion factors are applied to display values only; all internal computations remain in SI.

Table 22.3 Default display units

Quantity	SI Unit (default)	Imperial equivalent
Thrust	kN	lbf
Shaft Power	kW	hp
TSFC	kg/(kN·h)	lb/(lbf·h)
BSFC	kg/(kW·h)	lb/(hp·h)
Mass flow	kg/s	lb/s

Altitude	m	ft
Temperature	K	°R (for internal use only; GUI shows K)
Pressure	Pa / kPa	psia
Engine weight	kg	lb

23. Validation Methodology

23.1 Validation Framework

Validation in this project follows a two-track framework (established in §20.6): physics model validation against published reference data (Track A), and ML surrogate validation against held-out physics model data (Track B). The two tracks are never mixed. Validation tolerances are justified by reference confidence tier (§15 confidence system).

23.2 Physics Model Validation Reference Points

The following table defines the validation reference points for each engine. Every column is mandatory for each reference point. Columns marked "Estimated" indicate the reference value is from academic literature or derived calculation — not a manufacturer-certified measurement. Estimated reference values are used with wider tolerances.

Table 23.1 Physics model validation reference points (Track A) — selected representative conditions

Engine	Condition	H [m]	M	Throttle	AB	Ref. Thrust/Power	Ref. TSFC / BSFC	Source	Confidence	Tolerance
J79-GE-17	SLS Max Dry	0	0.0	1.0	OFF	52.8 kN (Ref. value)	~0.028 kg/(N·h) (Est.)	Jane's All the World's Aircraft; USAF AFSC TM	**MEDIUM**	±10%
J79-GE-17	SLS Max AB	0	0.0	1.0	ON	79.6 kN	~0.058 kg/(N·h) (Est.)	Jane's; USAF AFSC	**MEDIUM**	±15%
Atar 9C	SLS Max Dry	0	0.0	1.0	OFF	42.2 kN	Est.	Jane's; SNECMA product brochures (declassified)	**MEDIUM**	±10%

R-25-300	SLS Max Dry	0	0.0	1.0	OFF	40.2 kN (Estimated from DTIC)	Est.	DTIC declassified / Roskam Vol. 1 (estimated)	**MEDIUM**	±15%
CFM56-5B4	SLS Takeoff	0	0.0	1.0	N/A	120.0 kN	~0.013 kg/(N·h)	EASA TCDS E.042 (Type Certificate)	**HIGH**	±5%
CFM56-5B4	Cruise M0.82 / FL350	10,668	0.82	~0.85	N/A	~27 kN (Est. from SFC loop)	~0.016 kg/(N·h) (Est.)	Academic literature (Mattingly); EASA TCDS	**MEDIUM**	±10%
CF6-80C2B1F	SLS Takeoff	0	0.0	1.0	N/A	258.0 kN	~0.011 kg/(N·h)	GE Aviation data sheet (public); EASA TCDS E.003	**HIGH**	±5%
F100-PW-229	SLS Max Dry	0	0.0	1.0	OFF	79.2 kN	Est.	Jane's; Pratt & Whitney P&W229 press release (public)	**MEDIUM**	±10%
M88-2	SLS Max Dry	0	0.0	1.0	OFF	50.1 kN	Est.	Safran public brochure; Jane's	**MEDIUM**	±10%
Kaveri GTX-35VS	SLS Max Dry (Estimated)	0	0.0	1.0	OFF	~53.4 kN (ESTIMATED — academic model)	Not available	ADA/GTRE press releases; academic estimates — LOW confidence	**LOW**	±25%

PT6A-67D	SLS Max Takeoff Power	0	0.0	1.0	N/A	1,119 kW shaft power	~0.30 kg/(kW·h) (Est.)	Pratt & Whitney Canada PT6A series specifications; Jane's	**HIGH**	±5%
PW127G	SLS Max Takeoff Power	0	0.0	1.0	N/A	2,051 kW shaft power	~0.26 kg/(kW·h) (Est.)	EASA TCDS E.057; PWC PW100 brochure	**HIGH**	±5%
AE 2100D3	SLS Max Takeoff Power	0	0.0	1.0	N/A	3,458 kW shaft power	~0.24 kg/(kW·h) (Est.)	Rolls-Royce AE 2100 product data; Jane's; US MIL specs	**HIGH**	±5%

> **Validation Policy:** For HIGH confidence reference points (EASA TCDS, manufacturer data sheet), the tolerance is ±5%. For MEDIUM confidence (Jane's, DTIC, press releases), the tolerance is ±10–15%. For Kaveri (LOW confidence, estimated parameters), the tolerance is ±25% and results are marked ESTIMATED throughout. Validation is performed at model default component efficiencies; sensitivity to efficiency assumptions is documented separately.

23.3 ML Surrogate Validation (Track B)

ML surrogate validation uses the 15% held-out test split from the physics-model-generated dataset (§20.4). The following metrics are reported per engine per ML model type:

- R^2 on held-out test data
- RMSE, MAE, MAPE on test data
- R^2 vs training size curve (learning curve) to diagnose underfitting/overfitting
- Prediction time per query point (mean over 1,000 queries)
- Regional error map: identify (M, H) regions of high surrogate error

Track B validation results do not modify Track A physics model validation conclusions. Both tracks are reported in §27 Expected Outputs.

23.4 ISA Model Validation

The ISA model is validated against ICAO Doc 7488/3 tables at the nine altitude levels shown in Table 16.1 (§16.5). Acceptable agreement is within rounding error (agreement to 5 or more significant figures). This validation is automated as a unit test run before any engine computation.

24. Testing Strategy

24.1 Test Levels

Testing follows a four-level hierarchy corresponding to the development sequence defined in §25:

Table 24.1 Test level hierarchy

Level	Name	Scope	Tool	Pass Criterion
L1	Unit Tests	Individual MATLAB functions: ISA equations, nozzle check, fuel-air ratio, normalisation, constraint filter	MATLAB Unit Test Framework (`matlab.unittest`)	All assertions pass; no NaN returned for valid inputs
L2	Component Integration Tests	Full engine cycle: turbojet/turbofan/turboprop at known operating points; check energy balance	Script-based comparison vs hand-calculated reference values	Thrust and TSFC within ±1% of analytical hand-calculation
L3	System Integration Tests	DT Coordinator + DB + GUI interaction; end-to-end query-to-result flow; ML surrogate routing	MATLAB test scripts with database in test mode (separate test DB)	Correct Data_Type assigned; warning flags propagated to GUI; no orphan DB records
L4	Validation Tests (Track A/B)	Physics model vs reference data (§23 Table 23.1); ML model vs held-out physics data (§20.4)	Automated MATLAB validation scripts; manual cross-check of Table 23.1 points	Within tolerances defined in §23.2 and §20.5

24.2 Critical Unit Tests

The following unit tests are mandatory before any integration testing proceeds:

Table 24.2 Mandatory unit tests

Test ID	Function Tested	Test Case	Expected Output
UT-ISA-01	`ISA_model.m`	H = 11,000 m	T = 216.65 K, P = 22,632 Pa ± rounding
UT-ISA-02	`ISA_model.m`	H = −100 m (below sea level)	ERR-ATM-01 returned

UT-NOZ-01	Nozzle logic in `TJ_Model.m`	NPR = 2.5 > 1.854	Nozzle state = CHOKED; M_e = 1.0
UT-NOZ-02	Nozzle logic	NPR = 1.5 < 1.854	Nozzle state = UNCHOKED; M_e < 1.0
UT-NORM-01	Normalisation in `EngineSelector.m`	All engines have identical thrust value	All S_thrust = 0.5; tie message logged
UT-NORM-02	Normalisation	Two engines: x = [100, 200]	Higher-better: S = [0.0, 1.0]; lower-better: S = [1.0, 0.0]
UT-CONST-01	`apply_hard_constraints()`	All engines fail minimum thrust check	Empty feasible set; "No engine meets constraints" message
UT-CONST-02	`apply_hard_constraints()`	Single engine passes	Single-engine notice displayed; no ranking produced
UT-DB-01	`DB_Interface.m`	Parameterized INSERT then SELECT with Engine_ID = 1	Data retrieved matches inserted; foreign key enforced
UT-KAV-01	Kaveri model routing	Any Kaveri query	Data_Type = 'Simulated_Estimated'; Kaveri disclaimer banner visible

24.3 Regression Testing

After any code change to a physics model or database schema, regression tests compare current outputs against archived outputs from the last verified milestone. A regression failure (output change > 0.01%) must be investigated before further development proceeds.

24.4 Test Database

All testing uses a dedicated test database (`engine_test.db`) separate from the production database (`engine_analysis.db`). The test database contains all 11 engines but uses simplified reference values. The production database is never modified during automated testing.

25. Project Milestones

25.1 Overall Schedule

The project follows an 18-week development schedule (approximately one semester). The critical dependency rule is: **ML development does not begin until physics models have been individually validated (Track A)**. This prevents training surrogate models on unvalidated physics data.

> **Milestone Dependency:** Each milestone has one or more hard prerequisites. A milestone cannot begin until all its prerequisites are formally signed off. The project definition is the prerequisite for all implementation work.

25.2 Milestone Table

Table 25.1 Project milestones — Version 1.1 (18-week schedule)

MS #	Milestone	Week	Prerequisites	Deliverable	Pass Criterion
M01	Architecture Freeze	1	—	This document (v1.1, approved)	Document approved; no open architecture questions
M02	Database Schema Implementation	2–3	M01	`engine_analysis.db` (empty tables); `schema.sql`	All 11 tables created; FK constraints enforced; PRAGMA foreign_keys = ON verified
M03	Source Verification and DB Population	3–4	M02	Populated Engines, Manufacturers, References, Parameter_Provenance tables; confidence classification complete	All 11 engines populated; every parameter has provenance entry; Kaveri parameters flagged LOW
M04	ISA Model (Verified)	4	M01	`ISA_model.m` ; unit test suite UT-ISA-01, UT-ISA-02	ICAO Doc 7488/3 table agreement within rounding (Table 16.1)
M05	Turbojet Model (Coded)	4–5	M04	`TJ_Model.m` with validity checks and choked/unchoked nozzle logic	Unit tests UT-NOZ-01, UT-NOZ-02 pass; SLS thrust within ±15% of J79 reference

M06	Turbojet Model Validation (Track A)	5–6	M05	Validation report: J79, Atar 9C, R-25 at SLS reference points (Table 23.1)	All three turbojet engines within stated tolerances (±10–15%); results documented
M07	Turbofan Model (Coded and Validated)	6–8	M04, M06	`TF_Model.m`; validation vs CFM56-5B4, CF6-80C2, F100, M88, Kaveri	HBP engines within ±5% of EASA TCDS; LBP within ±10%; Kaveri flagged ESTIMATED
M08	Turboprop Model (Coded and Validated)	8–9	M04	`TP_Model.m`; static thrust test; validation vs PT6A-67D, PW127G, AE 2100D3	All three engines within ±5% at SLS; static thrust formula verified
M09	Database Population (Simulated Data)	9–10	M03, M06, M07, M08	Performance_Data table populated with L3 physics results; Simulation_Log entries created	All 11 engines × complete operating grids stored; Data_Type = 'Simulated' for physics; no NaN in valid-region results
M10	Digital Twin L1–L4 Integration	10–11	M09	`DT_Coordinator.m`, `DB_Interface.m`, `PerformanceMap.m`, `DTwin_L4.m`	End-to-end query-to-DB flow; system integration tests L3 pass; reference overlay working in GUI Tab 4
M11	ML Training and Evaluation	11–13	M09 (physics validated — mandatory)	Trained surrogates for all 11 engines × 4 model types; `ML_Surrogate.m`	Best model for each engine meets $R^2 \geq 0.95$ (target ≥ 0.99); results documented honestly
M12	Digital Twin L5 (ML Augmented)	13–14	M11	`DTwin_L5.m`; surrogate routing integrated in DT_Coordinator; ML fallback to physics	L5 query latency < 10 ms; ML/physics comparison plots generated; surrogate availability flag working
M13	Performance Comparison Module	14	M10, M12	Multi-engine overlay plots (Tab 4); architecture comparison charts; ranking display	All 11 engines can be overlaid; reference vs simulated vs ML clearly labelled

M14	Mission-Based Engine Selection	14–15	M09, M10	`EngineSelector.m`; Tab 6 GUI implementation; all unit tests UT-CONST-01, UT-CONST-02, UT-NORM-01, UT-NORM-02	Hard constraints applied first; empty/single-engine cases handled; tie case handled; results logged to DB
M15	GUI Integration	15–16	M13, M14	Complete `EngineAnalyser_v1.mlapp` with all 8 tabs; all mandatory warnings implemented	All mandatory warnings functional (Table 22.2); unit selector working; Kaveri disclaimer permanent
M16	System Integration and Regression Testing	16	M15	Complete test run: all unit tests + integration tests + regression tests	All UT-*, IT-* tests pass; no regression failures vs M10 baseline
M17	Final Validation and Documentation	17	M16	Validation report (Track A + Track B); user manual; project report draft	All Track A validation points documented with actual error %s; Track B R^2 documented; all limitations stated
M18	Project Submission	18	M17	Final PDF report; working MATLAB application; populated database; all source files in version-controlled repository	Complete deliverable package submitted per §27

25.3 Critical Path

The critical path is: M01 → M04 → M05 → M06 → M07 → M08 → M09 → M11 → M12 → M15 → M16 → M17 → M18. The physics model validation (M06, M07, M08) gates ML development (M11) — this is the most important sequencing constraint. Any slip in M06 or M07 directly delays M11 and all subsequent milestones.

26. Risks, Limitations, and Scope Boundaries

26.1 Explicit Scope Boundaries — What This System Does NOT Model

This section explicitly defines phenomena and features outside the project scope. These are not implementation gaps — they are design decisions reflecting the academic, steady-state, public-data nature of the project. Any result that appears to contradict these boundaries is an error, not a feature.

Table 26.1 Explicit out-of-scope phenomena

Category	Not Modelled	Rationale / Impact
Engine Dynamics	Transient engine behaviour (acceleration/deceleration)	Steady-state thermodynamic model only; all outputs represent equilibrium conditions
	Surge, stall, and rotating stall	No compressor maps; stability margins not computed
	Variable geometry scheduling (VGV, VSV)	Constant isentropic efficiencies assumed; off-design behaviour simplified
	Engine-start and windmill relight	No low-speed or low-pressure modelling
Thermodynamics	Compressor characteristic maps (β-map, N-line)	Single-point efficiencies used; performance map interpolation not available
	Turbine cooling flows (film/convection)	Cooling air bypasses turbine stage; not modelled; systematic error at high TIT
	Detailed combustion chemistry	Fuel-air ratio computed from energy balance only
	Afterburner thermochemistry (detailed)	Phase 1 afterburner treated as simple additional heat addition (if implemented)
Installation	Inlet distortion	Uniform inlet flow assumed; distortion-induced surge not modelled
	Bleed air extraction for airframe systems (air conditioning, pressurisation, avionics cooling)	Bleed air not deducted from compressor flow; optimistic shaft power output
	Nacelle installation drag, thrust reverser	No aircraft integration effects
Operations	Atmospheric turbulence and wind gradients	ISA standard day only; no atmospheric disturbances
	Engine degradation (fouling, erosion, tip clearance growth)	New-engine performance only; no time-in-service model
	Hot-section life and low-cycle fatigue	No structural model

Data	Classified military performance data	Public-domain only; DTIC/Jane's sources at MEDIUM confidence for military engines
	Proprietary manufacturer data (detailed performance decks)	Not publicly available; not used
	Real-time hardware interface (sensors, FADEC, test cell data)	Pure software system; no hardware-in-the-loop

The system is intended for: **"academic comparative performance analysis and engineering trend studies."** It is not suitable for flight planning, engine certification, airworthiness assessment, procurement decisions, or any safety-critical application.

26.2 Risk Register

Table 26.2 Project risk register

Risk ID	Risk Description	Probability	Impact	Mitigation
R01	Kaveri parameters unavailable from public sources; estimated values produce unrealistic performance	High	Medium	All Kaveri outputs permanently labelled ESTIMATED; wider tolerance (±25%); academic disclaimer mandatory
R02	Physics model for military engines (R-25, J79, M88) exceeds tolerance due to classified performance gaps	Medium	Medium	Use ±15% tolerance for military engines; document discrepancy; do not increase efficiency inputs to force agreement
R03	ML surrogate $R^2 < 0.95$ for some engines (unexpected nonlinearity or insufficient training data)	Low	Low	Report actual R^2; investigate feature engineering; document; do not force acceptance
R04	SQLite concurrent access issues during ML training + DB write	Low	Low	Single-threaded MATLAB operation; WAL journal mode enabled; test DB separate from production
R05	MATLAB toolbox version incompatibility (R2023b vs R2022a)	Low	Medium	Document minimum required MATLAB version; use only core toolbox functions available in R2022b+
R06	Turboprop static thrust formula produces large error vs reference data at M = 0	Medium	Medium	Actuator disc formula implemented; validated against PT6A-67D reference at SLS; blending function for $M \in [0, 0.05]$

R07	Mission selection always produces empty set for some preset profiles (constraint too tight for available engines)	Medium	Low	Empty-set handling implemented with informative message; mission presets pre-tested; custom profile option provided

26.3 Accuracy Caveats

All performance values produced by this system carry an intrinsic modelling uncertainty arising from: (1) calorically perfect gas assumption (systematic error at high temperatures); (2) single-point isentropic efficiency assumption (does not capture off-design degradation); (3) no turbine cooling model (overpredicts turbine work at high TIT); (4) constant component efficiency with throttle (not physical below ~70% throttle). These caveats apply universally and are irreducible within the project's scope. They do not invalidate the system's utility for comparative trend analysis.

27. Expected Outputs and Deliverables

27.1 Software Deliverables

Table 27.1 Software deliverables

ID	Deliverable	Format	Location	Notes
SW-01	MATLAB Application	`.mlapp` (App Designer)	Project root: `EngineAnalyser_v1.mlapp`	All 8 tabs functional; MATLAB R2023b+ required
SW-02	Physics model functions	`.m` files	`/models/`: `ISA_model.m`, `TJ_Model.m`, `TF_Model.m`, `TP_Model.m`, `DT_Coordinator.m`	Fully commented; every function has an associated unit test
SW-03	ML surrogate models	`.mat` files (MATLAB model objects)	`/ml_models/`: one file per engine per model type	11 engines × 4 model types = 44 trained models (best per engine identified)
SW-04	Database	SQLite (`.db`)	`engine_analysis.db` (production); `engine_test.db` (test)	Both files provided; schema version documented in metadata table
SW-05	Database schema script	SQL (`.sql`)	`schema.sql`	Fully reproducible; creates all tables with constraints from a blank database

SW-06	Unit and integration test suite	`.m` test files	`/tests/`: all UT-* and IT-* tests	Run via `runtests('tests/')` in MATLAB

27.2 Documentation Deliverables

Table 27.2 Documentation deliverables

ID	Deliverable	Format	Notes
DOC-01	Project Definition & Architecture Document v1.1	PDF	This document; supersedes v1.0
DOC-02	Validation Report	PDF	Track A (physics vs reference) and Track B (ML vs physics) results; actual error % for every reference point in Table 23.1
DOC-03	User Manual	PDF	GUI operation guide; all 8 tabs; warning interpretation; data export instructions
DOC-04	Final B.Tech Project Report	PDF	Academic report per university format; cites this specification document throughout

27.3 Minimum Acceptance Criteria for Project Completion

1. All 11 engines modelled across their valid operating envelopes (as defined in Table 18.5).
2. All physics models validated at sea-level static conditions (Table 23.1) with results documented.
3. ML surrogates trained for all 11 engines; actual R^2 values reported (not targets).
4. Mission-based selection working for all 6 preset missions with correct hard-constraint filtering.
5. GUI displaying all mandatory warnings per Table 22.2.
6. Kaveri disclaimer permanently visible in all Kaveri computations.
7. All ML validation using physics-model-generated held-out data (Track B only) — never reference engine data.
8. Database populated with all reference and simulated data, provenance tracked for every parameter.
9. Complete test suite passing (all UT-*, IT-* tests).
10. DOC-01 through DOC-04 submitted in final form.

28. Version 1.1 Change Log

This section documents all corrections, additions, and clarifications made in Version 1.1 relative to Version 1.0 (September 2026). Changes are numbered CR-01 through CR-20 corresponding to the 20 revision areas identified in the review.

Table 28.1 Version 1.1 change log

CR #	Section(s) Changed	Problem Identified in v1.0	Correction Made in v1.1	Reason	Impact on Implementation
CR-01	§9, §14, §15	Engine parameter numerical values presented without source classification; no distinction between manufacturer-confirmed, type-certificate, academic, calculated, or estimated values	Every engine parameter in §9 and §14 is now classified by source tier (HIGH / MEDIUM / LOW) using a six-tier system. Parameter_Provenance table added to schema (§11).	Prevents confusion between confirmed performance specifications and estimated values; enables honest validation tolerance setting	DB must be populated with provenance data per parameter; Parameter_Provenance table must be implemented and linked to every external value
CR-02	§9	Klimov TV3-117VM incorrectly classified as a turboprop. TV3-117 is a turboshaft helicopter engine family, not a fixed-wing turboprop.	TV3-117VM removed and replaced with Pratt & Whitney Canada PW127G (EASA TCDS E.057, 2,051 kW, application: ATR 72-600). Engine parameters updated with HIGH confidence sources.	Technical error: including a turboshaft as a turboprop representative would invalidate the turboprop architecture model comparisons	Physics model applies to PW127G with correct shaft/gearbox parameters; Turboprop ML training grid updated with PW127G specifications

CR-03	§17.2	Nozzle expansion equations in turbojet model did not handle the choked nozzle case. Exit pressure was always set to P_a regardless of NPR, which is physically incorrect at supersonic operating conditions of J79, Atar 9C, R-25.	Explicit NPR check added (§17.2.5). NPR_crit = 1.854 for γ_g = 1.33. Choked and unchoked cases handled with separate equations. Nozzle state logged to Simulation_Log.	Without this correction, thrust would be underestimated at high-altitude supersonic conditions and pressure thrust term would be zero when it should be non-zero	TJ_Model.m must implement the NPR check before nozzle exit calculation; nozzle state flag must be stored
CR-04	§17.3	Turbofan model was not explicitly designated as separate-exhaust or mixed-exhaust, creating ambiguity for MATLAB implementation.	Phase 1 turbofan model explicitly designated as SEPARATE-EXHAUST with two independent nozzles (core at Station 8, bypass at Station 18). Model applies to both HBP and LBP turbofans with different architecture parameters.	Separate-exhaust is simpler, analytically transparent, and appropriate for B.Tech scope; mixed-exhaust deferred to Phase 2	TF_Model.m must implement separate core and bypass nozzle paths with independent NPR checks; no mixing plane calculation required in Phase 1
CR-05	§17.4	Turboprop model did not include gearbox efficiency (η_gb) in the power transmission chain; ESHP formula was incomplete; static thrust (M = 0) produced division by zero.	Gearbox efficiency (η_gb = 0.98) added as explicit parameter in turboprop model. ESHP formula corrected. Static thrust handled via actuator disc theory with propeller diameter input. Blending function for M ∈ [0, 0.05].	Omitting gearbox efficiency overestimates propeller shaft power; static thrust singularity would crash the MATLAB model	TP_Model.m must implement η_gb, actuator disc static thrust, and blending function; propeller diameter must be stored in DB for all three turboprop engines
CR-06	§16	Altitude convention (geometric vs geopotential) was not stated; ISA equations used without clarification of altitude type assumption.	Geopotential altitude H defined explicitly (§16.1). Conversion formula from geometric to geopotential provided. 0.16% approximation error documented and accepted.	Ambiguity in altitude convention leads to small but systematic errors in ISA calculations; must be resolved before implementation	GUI must accept geometric altitude as user input and convert to geopotential internally; ISA_model.m must document altitude convention in function header

CR-07	§11	References system traced each engine to one row in References table; this is insufficient for per-parameter traceability required for an honest validation report.	New table Parameter_Provenance added to schema (§11.9). Each row links one parameter of one engine to one reference source with confidence tier and value classification. FK references both Engines and References tables.	Without per-parameter provenance, validation report cannot identify which specific values are manufacturer-confirmed vs estimated	Parameter_Provenance must be populated for all numerical parameters of all 11 engines before validation begins; adds approximately 150–200 rows to the DB
CR-08	§11.8, §14	Performance_Data table did not clearly distinguish Reference / Simulated / ML_Predicted data types, risking conflation during validation comparisons.	Data_Type field with CHECK constraint ('Reference', 'Simulated', 'ML_Predicted', 'Simulated_Estimated') enforced in Performance_Data. Prediction_Source field added. Kaveri-specific data type 'Simulated_Estimated' introduced.	Mixing reference and simulated data in validation comparisons would produce meaningless validation metrics	DB_Interface.m must correctly assign Data_Type based on computation path; ML_Surrogate.m must set Prediction_Source = 'ML'; physics models set Prediction_Source = 'Physics'
CR-09	§19	Digital Twin described without academic scope clarification; language implied capabilities (real-time, hardware-connected) not achievable in this project.	Academic scope disclaimer added to §19 explicitly defining what the DT is and what it is NOT. Five-level structure preserved but scope boundaries clarified.	Overstatement of DT capability would mislead the academic examiner and create false implementation expectations	Documentation must consistently refer to "academic steady-state software Digital Twin" rather than implying real-time or hardware capabilities

CR-10	§20	ML validation used ML vs reference data comparison, which conflates physics model accuracy and ML surrogate accuracy. Single cross-architecture ML model was proposed without scientific justification.	ML validation explicitly restricted to Track B: ML vs held-out physics-model data only. Architecture-specific surrogates required (one per engine architecture class). Separation of Track A and Track B validation formalized in §20.6.	A single cross-architecture ML model trained on turbojet, turbofan, and turboprop data simultaneously is scientifically unjustifiable given incompatible Mach ranges and output variables	ML_Surrogate.m must maintain separate model objects per engine; no cross-architecture training; Track B metrics must not be presented as physics model accuracy
CR-11	§20.1	Mach range for ML training data was uniform across all engine types (9 points 0.0–2.5), generating physically invalid data points for turboprop and high-BPR turbofan engines.	Mach ranges made engine-type-specific in ML training grids (Table 20.1). Invalid-condition filtering applied before any data storage. Turboprop Mach range capped at 0.7; HBP turbofan at 0.92.	Including M = 2.5 for turboprop in training data would train the surrogate on physically impossible operating conditions, producing incorrect performance predictions	Training data generation script must apply validity checks (§18) before storing any data point; architecture-specific grid parameters must be parameterized in the generation script
CR-12	§20.5	$R^2 \geq 0.99$ stated as a guaranteed acceptance criterion. This is not achievable in general and creates pressure to manipulate datasets to force artificial compliance.	$R^2 \geq 0.99$ reclassified as a target, not a guaranteed threshold. Minimum acceptable $R^2 = 0.95$. If below threshold, actual result is documented and root causes investigated. Dataset manipulation prohibited.	Presenting an aspirational metric as a guaranteed threshold creates academic dishonesty risk and hides genuine model limitations	Validation report template must include "Target", "Actual", and "Assessment" columns; results below target must be accompanied by investigation notes

CR-13	§21	Engine selection algorithm applied weighted scoring without prior hard-constraint filtering. An engine with a fatal constraint violation (e.g., insufficient thrust for the mission) could win the ranking through high overall score.	Two-stage algorithm introduced: Stage 1 applies hard constraints (eliminates non-feasible engines); Stage 2 applies weighted scoring to feasible set only. Empty and single-engine feasible set cases explicitly handled.	Allowing constraint-violating engines to compete in weighted scoring is a fundamental algorithm design error that could recommend an engine incapable of performing the mission	EngineSelector.m must implement apply_hard_constraints() as a separate function called before scoring; GUI Tab 6 must display elimination reasons
CR-14	§21.2	Normalisation formula did not handle the degenerate case x_max = x_min (all feasible engines have identical performance on a metric), causing division by zero.	Degenerate case explicitly handled: when x_max = x_min, output S_i = 0.5 for all engines for that metric, with a GUI log message indicating no discrimination.	Division by zero would crash EngineSelector.m at runtime; degenerate case is realistic when only one engine passes hard constraints	EngineSelector.m normalisation function must check x_max == x_min before computing score; error-handling implemented as unit test UT-NORM-01
CR-15	§21.3	Thrust-to-weight ratio definition was ambiguous; it was unclear whether engine weight or aircraft MTOW was used.	TWR explicitly defined as engine-level ratio: F_rated / (W_engine_dry × g_0). Aircraft-level TWR explicitly excluded from Phase 1. Aircraft mass and drag not modelled.	Using aircraft MTOW for TWR requires aircraft mass data not available in the database; engine-level TWR is computable from existing data and is consistent	Engines table dry weight field must be populated for all 11 engines; TWR computed internally from Engines table; no aircraft mass fields required in Phase 1

CR-16	§25	ML development milestone was not sequenced after physics model validation; ML could theoretically begin before physics models were individually validated, producing surrogates trained on unvalidated data.	M11 (ML Training) prerequisite set to M09 with explicit note "physics validated — mandatory". M09 is only reachable after M06 (turbojet validation), M07 (turbofan validation), and M08 (turboprop validation) are complete.	Training ML surrogates on unvalidated physics model outputs would render Track B validation meaningless and produce potentially incorrect performance predictions	Project management: no ML code may be committed to version control until M06, M07, and M08 sign-offs are documented
CR-17	§22	GUI specification did not include mandatory warnings for model validity, estimated data, steady-state assumption, or ML fallback. Unit display was not specified.	Mandatory warning and indicator set defined in Table 22.2. Eight required warnings specified with colours, locations, and messages. Units table (Table 22.3) added. Steady-state disclaimer made permanent in Tab 2 footer.	Without explicit warnings, users may interpret estimated Kaveri data as manufacturer-confirmed, or steady-state results as transient predictions	App Designer callbacks must check and display all eight conditions listed in Table 22.2; warnings must not be suppressible by user
CR-18	§18 (new)	No section in v1.0 defined the complete set of model validity checks; FR-07 required validity checks but they were not specified in detail.	New §18 (Model Validity Checks) added with four categories: input range, component efficiency, thermodynamic consistency, and nozzle condition. 30+ individual checks defined with error codes, severity, and actions.	Without explicit validity checks, MATLAB models may silently produce NaN, Inf, or physically impossible results that propagate through the system undetected	DT_Coordinator.m must call validity check function as first action before any model evaluation; all error codes must be logged

CR-19	§26	Limitations section in v1.0 was generic. Several important out-of-scope phenomena (compressor maps, turbine cooling, bleed air, installation effects, degradation) were not explicitly named.	§26 expanded with explicit table of out-of-scope phenomena (Table 26.1) organized by category: Engine Dynamics, Thermodynamics, Installation, Operations, Data. Statement of intended use added.	Omitting specific limitations creates ambiguity about system capability and risks overstatement of results in the project report and academic examination	Project report §26 language must be used verbatim in the academic report's limitations section; no limitations not listed here may be claimed as "modelled"
CR-20	§23, §28, §29	v1.0 did not have a Change Log or an Implementation Freeze Checklist. Validation reference points were not specified with source confidence or tolerances.	§28 (Change Log, this section) and §29 (Implementation Freeze Checklist) added. §23 validation table extended with Source and Confidence and Tolerance columns. Track A/B distinction formalized.	Without a Change Log, v1.0 vs v1.1 differences cannot be formally tracked. Without a Freeze Checklist, the transition from design to implementation lacks a formal gate.	DOC-01 (this document) supersedes v1.0 as the master specification. Freeze Checklist (§29) must be completed and signed before MATLAB coding begins.

29. Implementation Freeze Checklist

This checklist must be completed and formally signed off before any MATLAB coding begins. Each item must be verified against this document (v1.1). A "PASS" requires positive verification; "N/A" is only permitted for items explicitly excluded by scope. Any "FAIL" must be resolved by updating this document and re-issuing before implementation starts.

29.1 Architecture and Scope Checklist

Table 29.1 Architecture and scope verification

Item	Check	Reference	Status
FC-ARCH-01	Project scope is defined as academic comparative analysis only — no certification, no flight-safety claims	§1, §26	☐
FC-ARCH-02	Three engine architecture groups confirmed: turbojet (3), turbofan (5), turboprop (3) = 11 engines total	§6, §9	☐
FC-ARCH-03	PW127G (not TV3-117VM) confirmed as third turboprop representative	§9, CR-02	☐

FC-ARCH-04	All 11 engine architecture classifications verified (not turboshaft, not turbojet-with-augmentor misclassified as turbofan)	§6, §9	☐
FC-ARCH-05	Digital Twin scope confirmed as academic steady-state software DT — no real-time, no hardware-in-the-loop	§19	☐

29.2 Engine Data Checklist

Table 29.2 Engine data verification

Item	Check	Reference	Status
FC-DATA-01	Every engine parameter in the database has a source classification (HIGH / MEDIUM / LOW / Estimated / Calculated) in Parameter_Provenance	§11.9, §15	☐
FC-DATA-02	Kaveri GTX-35VS parameters marked LOW confidence; disclaimer applied to all Kaveri outputs	§9, §18.5	☐
FC-DATA-03	No parameter value was fabricated — all values trace to a cited public-domain source or are explicitly marked Estimated	§15, §23	☐
FC-DATA-04	Propeller diameter stored for all three turboprop engines (PT6A-67D, PW127G, AE 2100D3)	§9, §17.4.3	☐
FC-DATA-05	Engine dry weight stored for all 11 engines (required for TWR computation)	§21.3	☐

29.3 Physics Model Checklist

Table 29.3 Physics model implementation requirements

Item	Check	Reference	Status
FC-PHY-01	ISA model uses geopotential altitude; GUI converts geometric input to geopotential internally	§16.1	☐
FC-PHY-02	Turbojet model implements choked/unchoked nozzle logic with NPR_crit = 1.854 (γ_g = 1.33)	§17.2.5	☐
FC-PHY-03	Turbofan model is confirmed SEPARATE-EXHAUST with independent core (St.8) and bypass (St.18) nozzles	§17.3, CR-04	☐
FC-PHY-04	Turboprop model includes gearbox efficiency η_gb = 0.98 in power transmission chain	§17.4.3, CR-05	☐
FC-PHY-05	Turboprop static thrust (M = 0) uses actuator disc theory — no division by V_0	§17.4.3	☐

FC-PHY-06	All engine models use consistent station numbering: St.0, St.2, St.3, St.4, St.5, St.7 (TP), St.8, St.18 (TF)	§17.1	☐
FC-PHY-07	All model outputs use SI units internally; unit conversion only in GUI display layer	§22.4	☐

29.4 Model Validity Checks Checklist

Table 29.4 Validity check implementation requirements

Item	Check	Reference	Status
FC-VAL-01	DT_Coordinator calls validity check function before any engine model evaluation	§18, §19.2	☐
FC-VAL-02	All 8 input range checks implemented with correct error codes (Table 18.1)	§18.1	☐
FC-VAL-03	All 7 component efficiency range checks implemented (Table 18.2)	§18.2	☐
FC-VAL-04	All 9 thermodynamic consistency checks implemented (Table 18.3)	§18.3	☐
FC-VAL-05	NaN / Inf / complex result check implemented as final output validation step	§18.3 (ERR-NAN-01)	☐
FC-VAL-06	Unit tests UT-ISA-01, UT-ISA-02, UT-NOZ-01, UT-NOZ-02 all pass	§24.2	☐

29.5 Database Checklist

Table 29.5 Database implementation requirements

Item	Check	Reference	Status
FC-DB-01	All 11 SQLite tables created per §11 schema with all stated constraints	§11	☐
FC-DB-02	PRAGMA foreign_keys = ON verified active at connection time	§11	☐
FC-DB-03	Parameter_Provenance table implemented and populated for all 11 engines	§11.9, CR-07	☐
FC-DB-04	Performance_Data Data_Type CHECK constraint implemented with all four valid values	§11.8, CR-08	☐

FC-DB-05	All SQL queries in DB_Interface.m use parameterised queries (no string concatenation with user input)	§11	☐
FC-DB-06	Unit test UT-DB-01 passes	§24.2	☐
FC-DB-07	Production and test databases are separate files; production DB not modified during testing	§24.4	☐

29.6 Machine Learning Checklist

Table 29.6 ML architecture requirements

Item	Check	Reference	Status
FC-ML-01	ML training does not begin until M06, M07, M08 sign-offs are documented	§25, CR-16	☐
FC-ML-02	Separate surrogate models trained per engine architecture class (no cross-architecture model)	§20.2, CR-10	☐
FC-ML-03	Training data split is 70/15/15 (train/validation/test) with stratification by (altitude bin, Mach bin)	§20.4	☐
FC-ML-04	ML training data generated only within valid Mach and altitude ranges per engine type (Table 18.5, Table 20.1)	§18, §20.1, CR-11	☐
FC-ML-05	ML acceptance criteria treat $R^2 \geq 0.99$ as target not guarantee; actual R^2 documented regardless	§20.5, CR-12	☐
FC-ML-06	Track B validation (ML vs physics) and Track A validation (physics vs reference) kept strictly separate in the validation report	§20.6, §23.1	☐

29.7 Engine Selection and GUI Checklist

Table 29.7 Engine selection and GUI requirements

Item	Check	Reference	Status
FC-SEL-01	apply_hard_constraints() called before scoring in all mission selection runs	§21.1, CR-13	☐
FC-SEL-02	Empty feasible set handled with informative message (no crash, no ranking)	§21.1	☐
FC-SEL-03	Single-engine feasible set handled with special notice (performance report shown, no ranking)	§21.1	☐

FC-SEL-04	x_max = x_min degenerate normalisation case returns S_i = 0.5 for all engines	§21.2, CR-14	☐
FC-SEL-05	TWR uses engine dry weight from Engines table (not aircraft MTOW)	§21.3, CR-15	☐
FC-GUI-01	All 8 mandatory warnings from Table 22.2 are implemented and non-suppressible	§22.3, CR-17	☐
FC-GUI-02	Kaveri disclaimer "ESTIMATED / ACADEMIC MODEL — NOT MANUFACTURER-CONFIRMED PERFORMANCE" permanently displayed for all Kaveri computations	§18.5, §22.3	☐
FC-GUI-03	Steady-state model disclaimer permanently displayed in Tab 2 footer	§22.3	☐
FC-GUI-04	All input sliders display units and valid range annotations	§22.3	☐

> **Freeze Sign-off:** This Implementation Freeze Checklist is considered complete when all items above are marked PASS (or N/A where explicitly permitted). The project supervisor or academic advisor should review and countersign before implementation begins. Any item left as FAIL or blank is an unresolved design issue that must be addressed in a document revision before proceeding.

Sign-off	Name	Role	Date	Signature
Author		B.Tech Student		
Supervisor		Project Guide		